

BFS Traversal

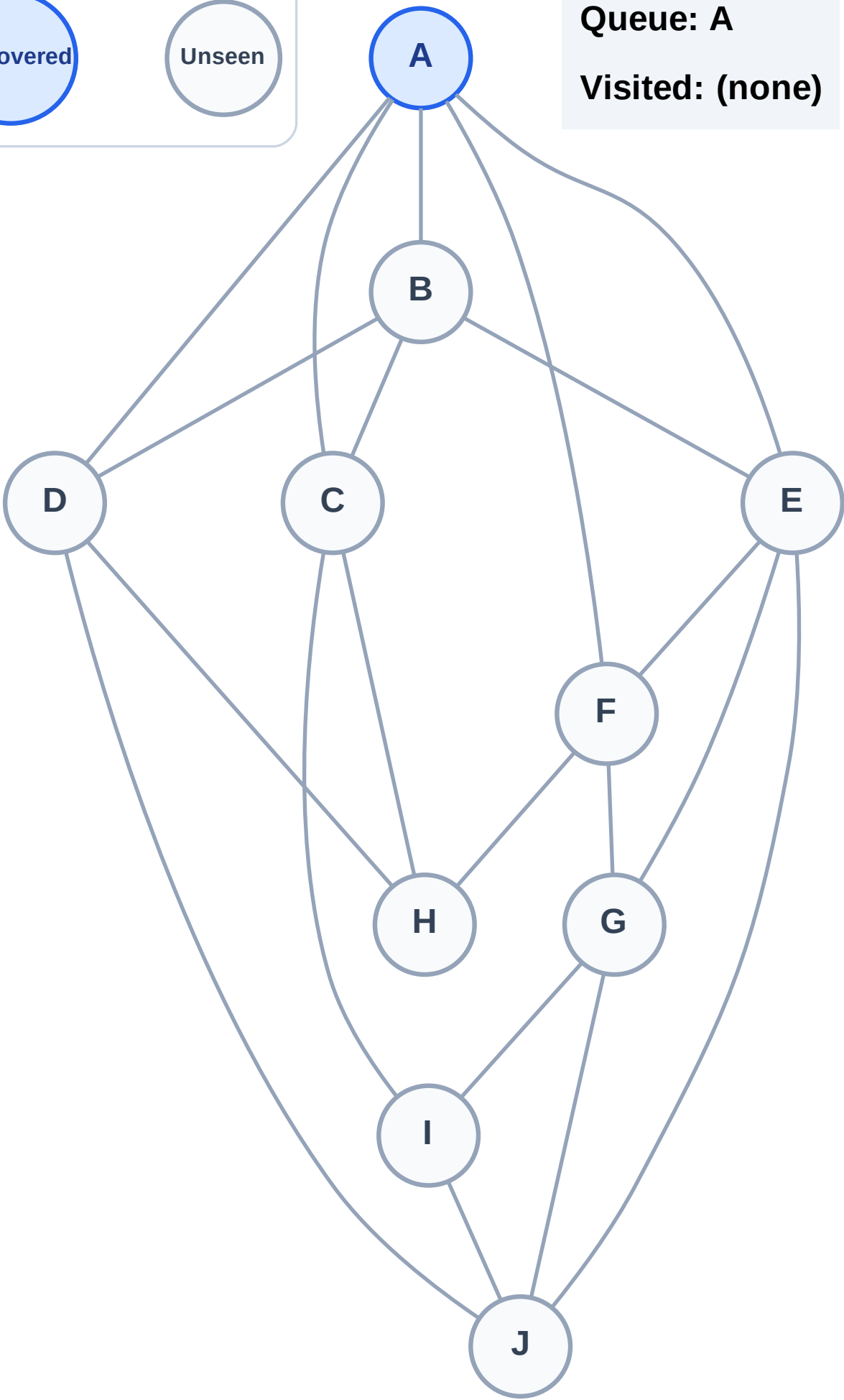
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

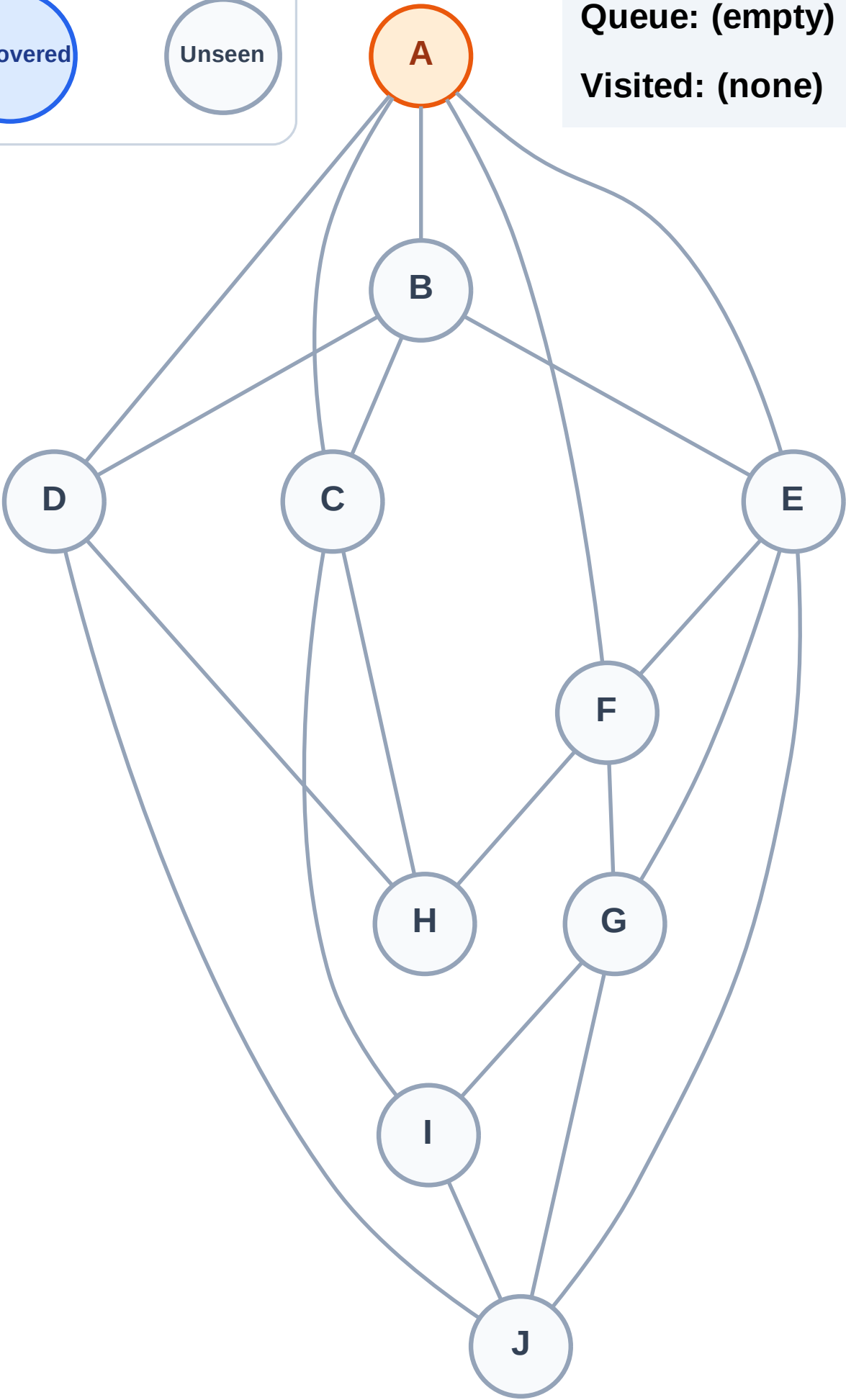
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

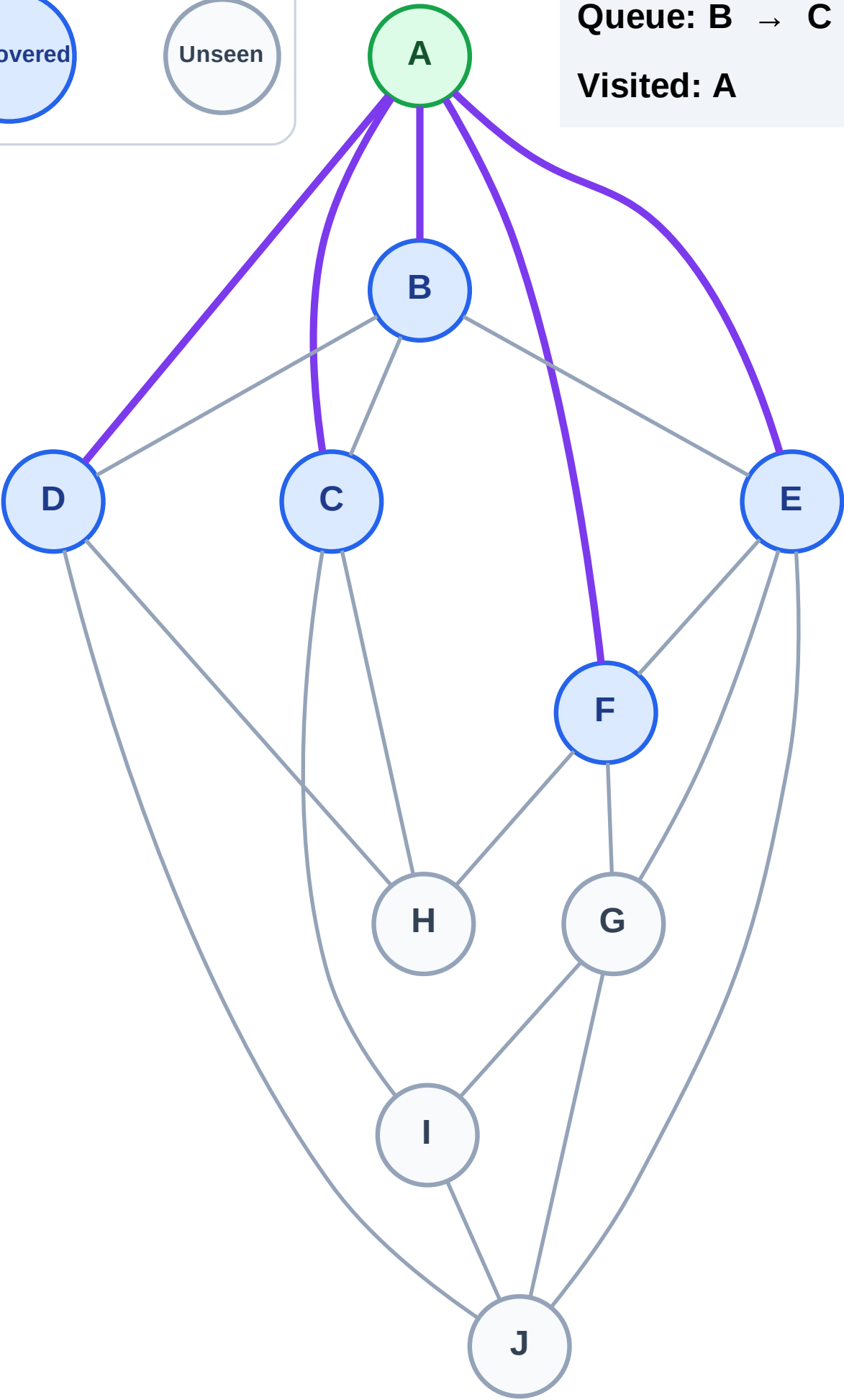
Step 3: Discover B, C, D, E, F from A

Legend



Queue: B → C → D → E → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

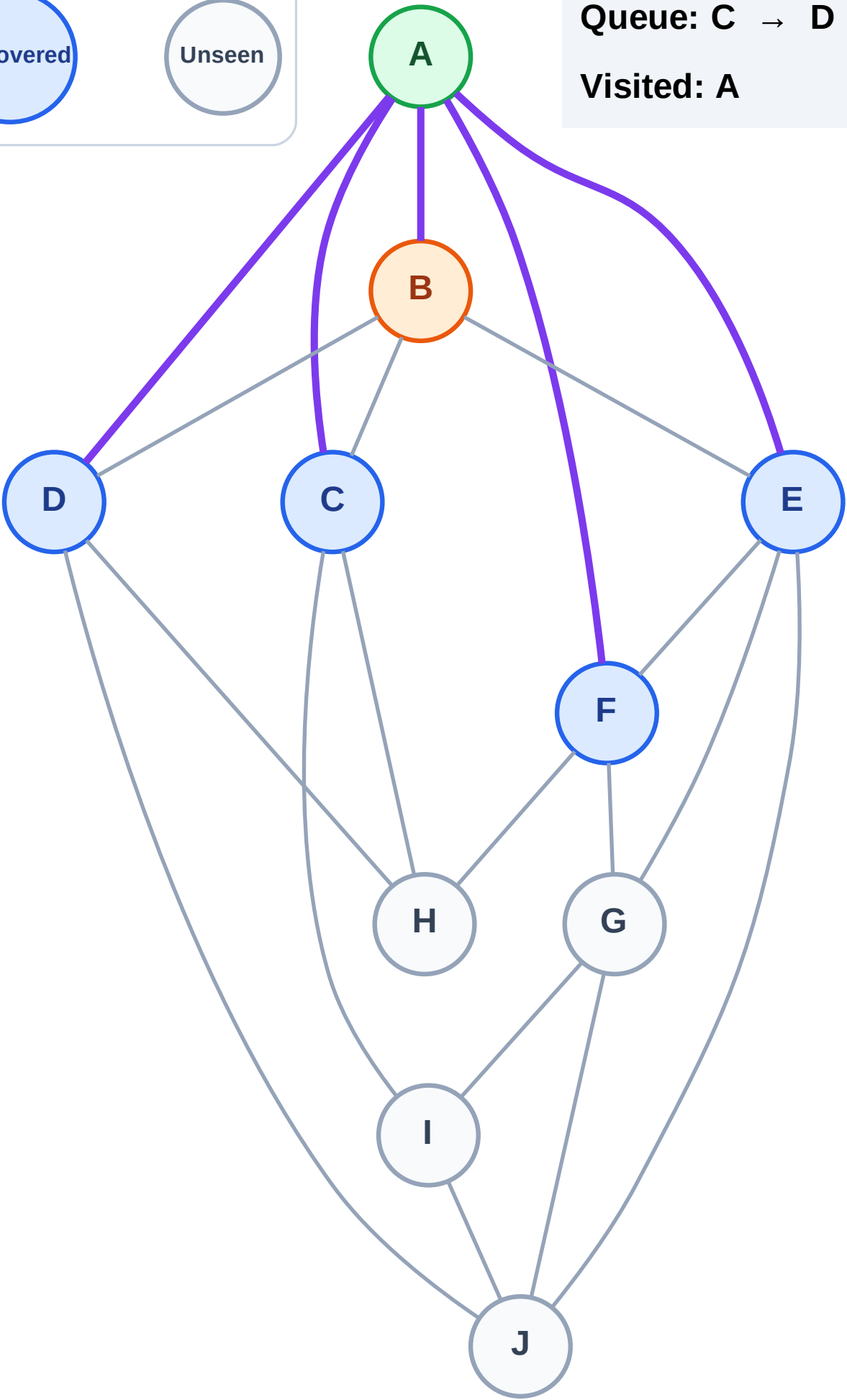
Processing

Discovered

Unseen

Queue: C → D → E → F

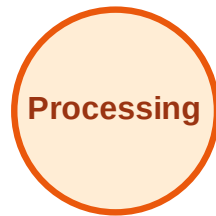
Visited: A



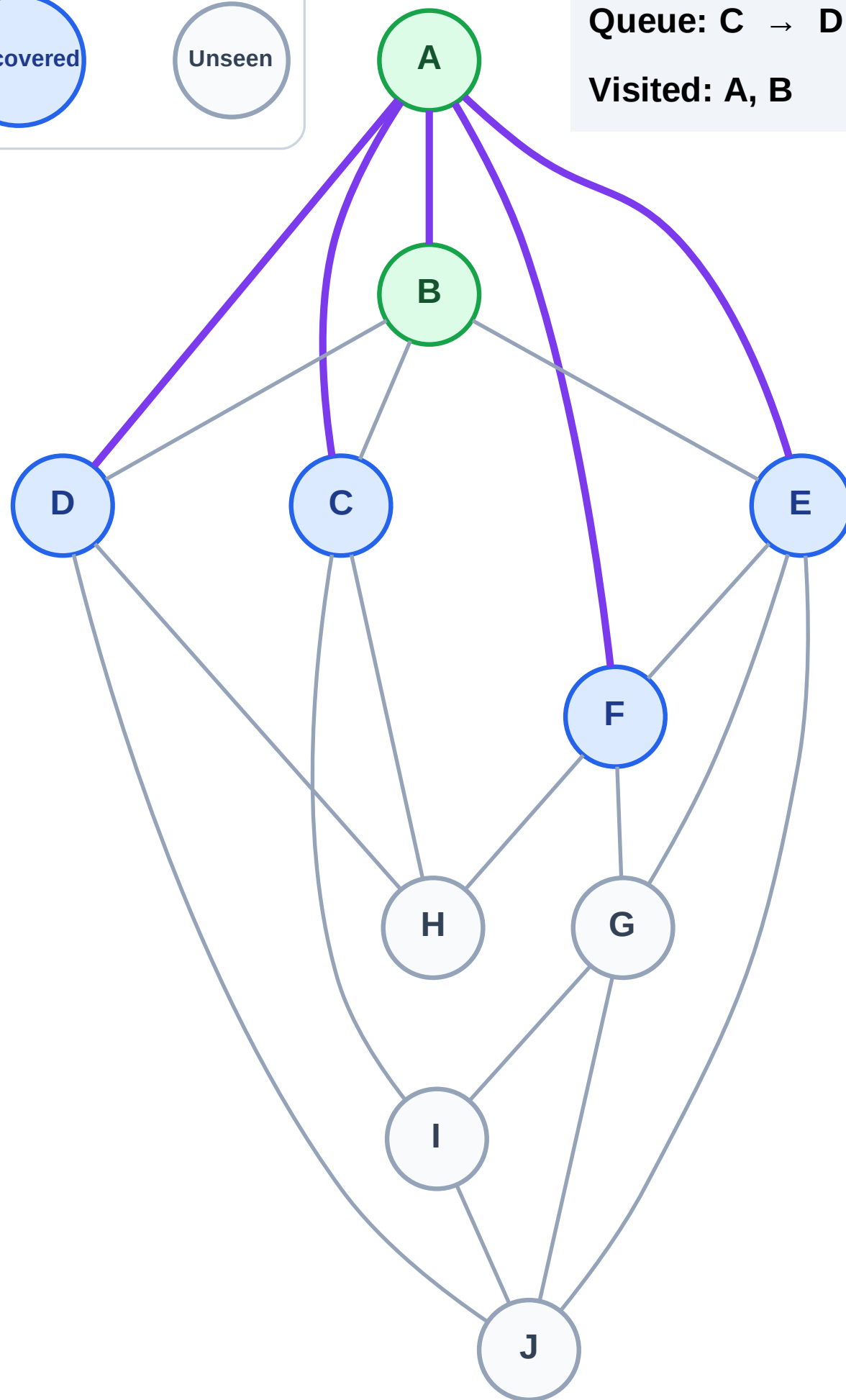
BFS Traversal

Step 5: No new neighbors from B

Legend



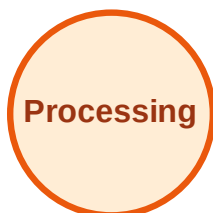
Queue: C → D → E → F
Visited: A, B



BFS Traversal

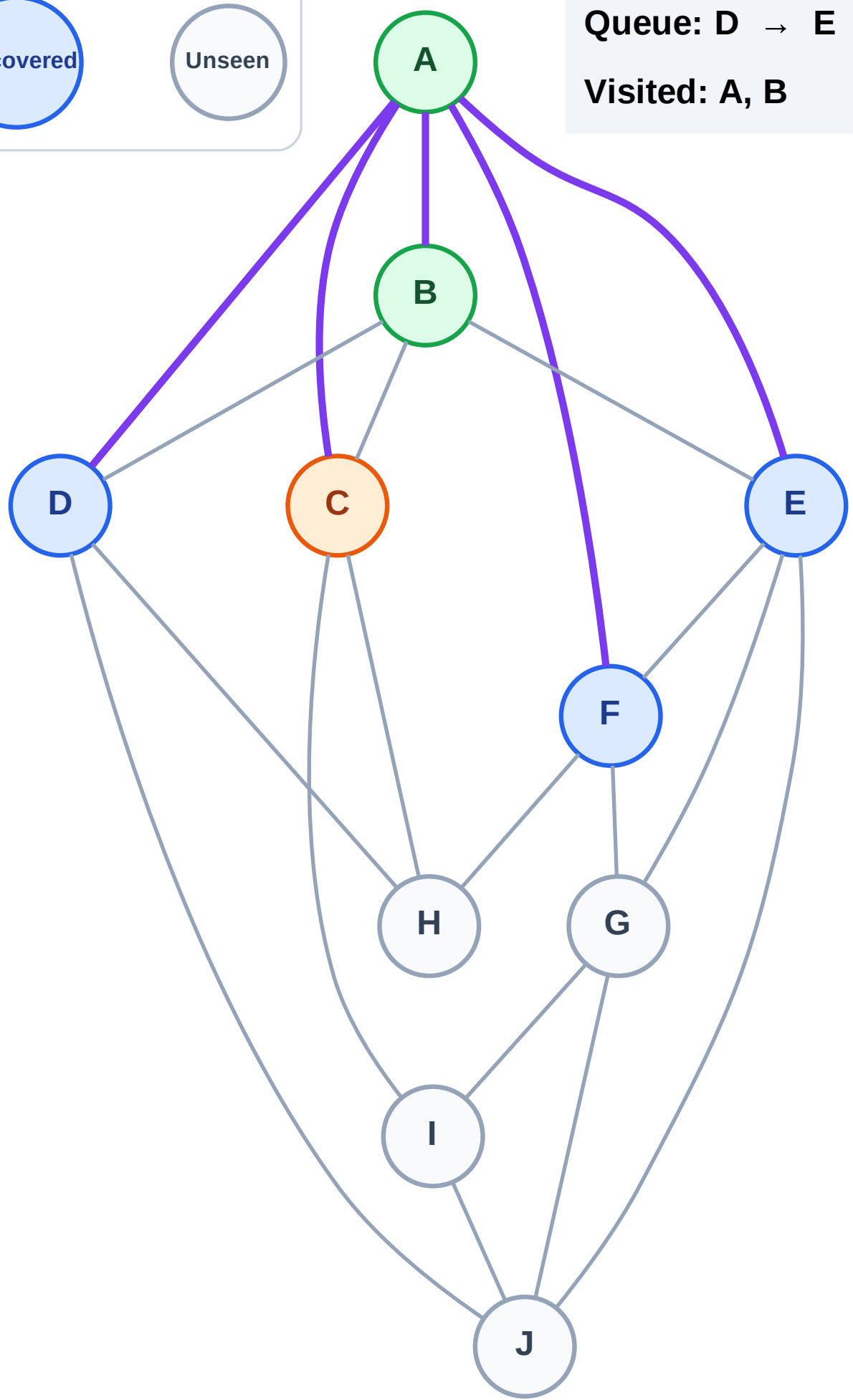
Step 6: Process node C

Legend



Queue: D → E → F

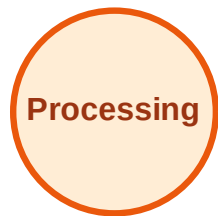
Visited: A, B



BFS Traversal

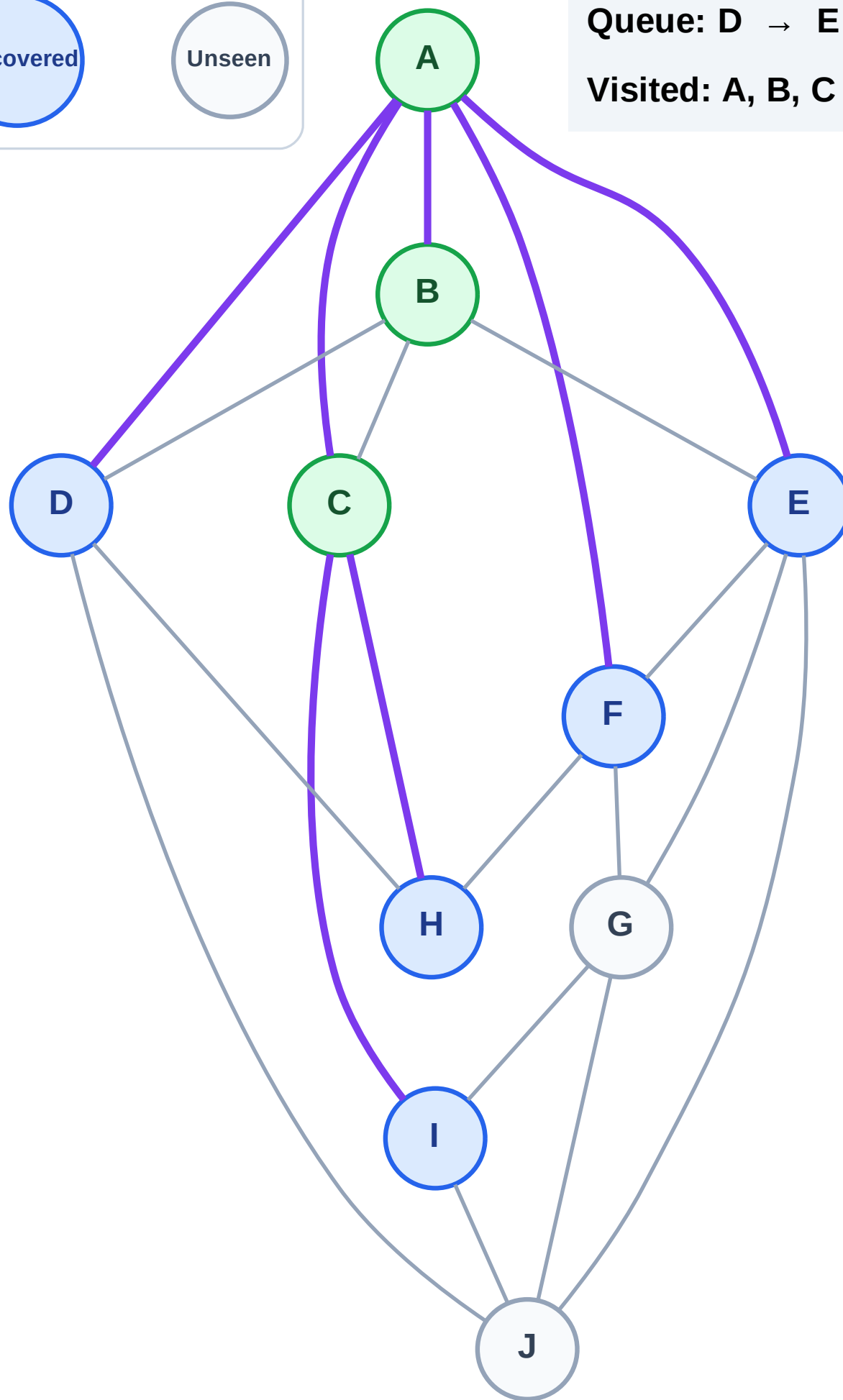
Step 7: Discover H, I from C

Legend



Queue: D → E → F → H → I

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

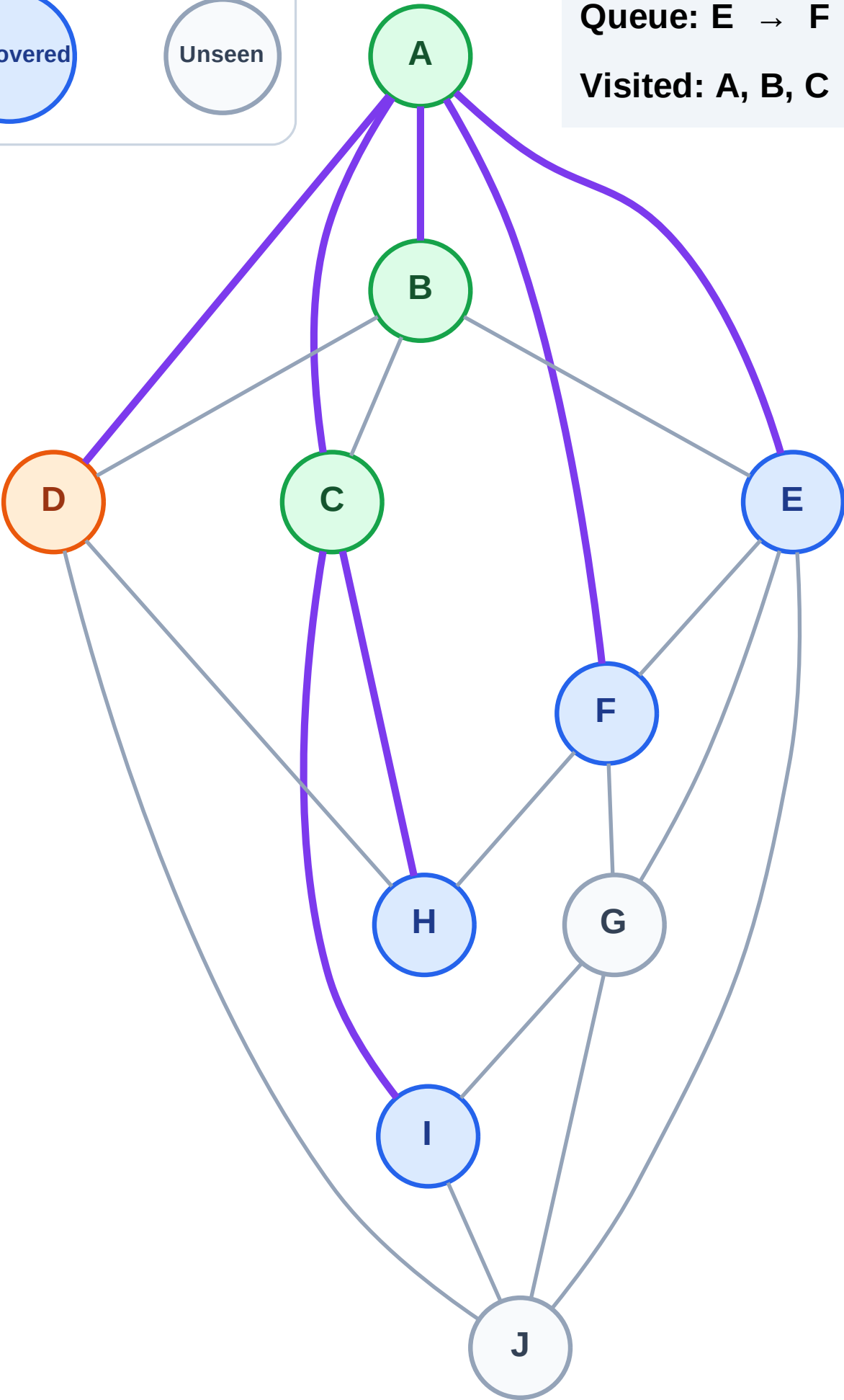
Processing

Discovered

Unseen

Queue: E → F → H → I

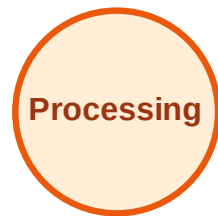
Visited: A, B, C



BFS Traversal

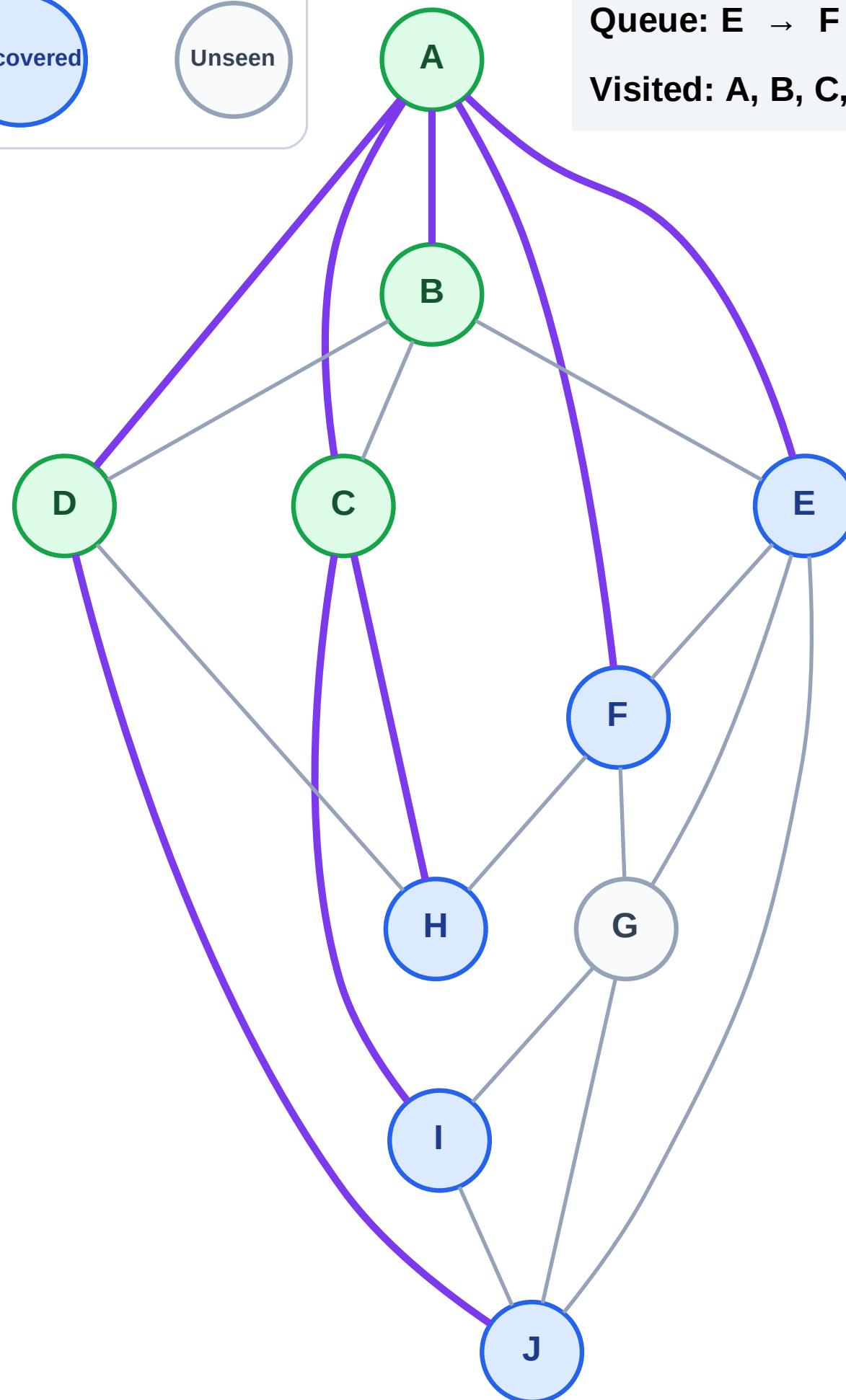
Step 9: Discover J from D

Legend



Queue: E → F → H → I → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

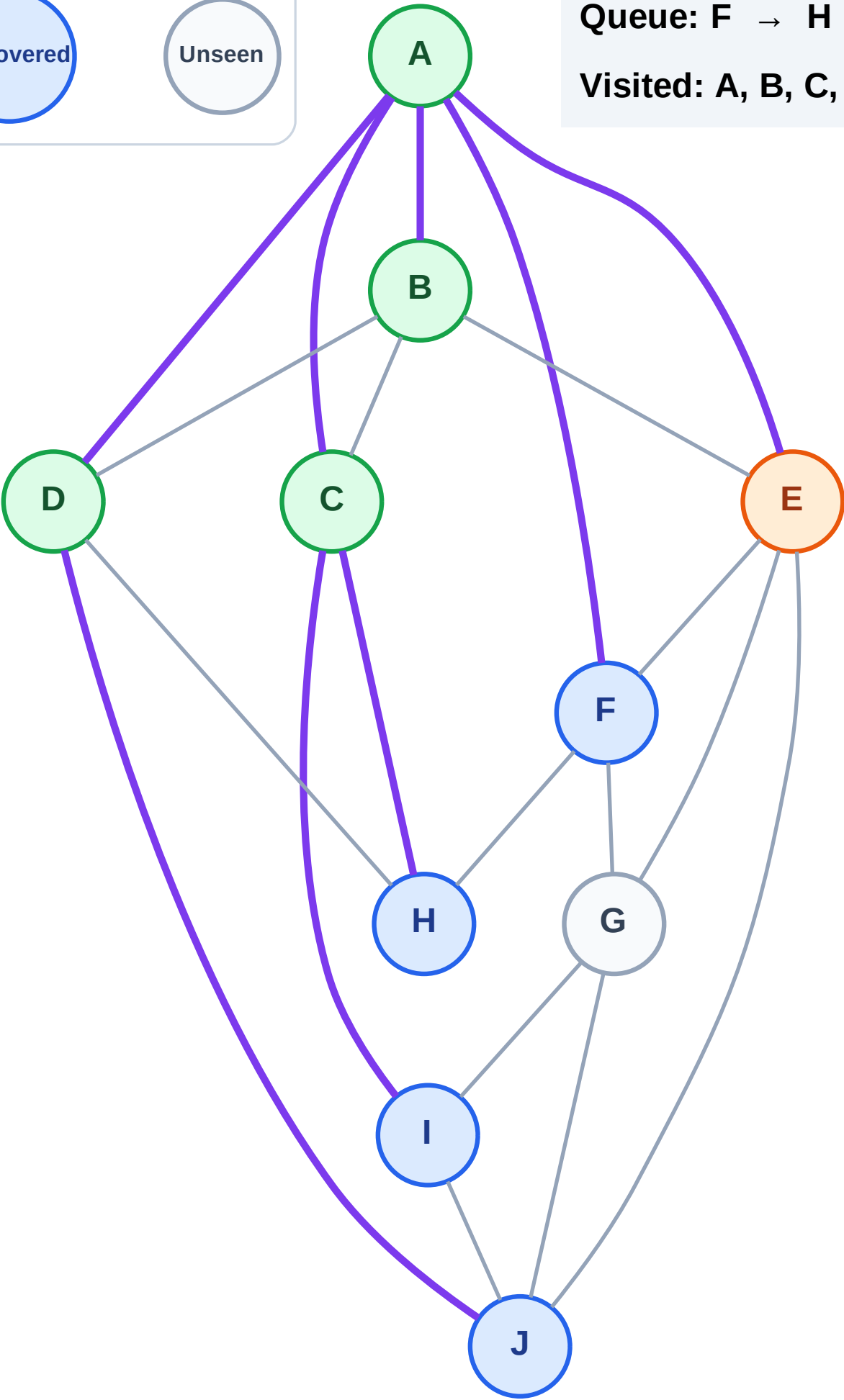
Processing

Discovered

Unseen

Queue: F → H → I → J

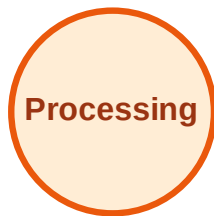
Visited: A, B, C, D



BFS Traversal

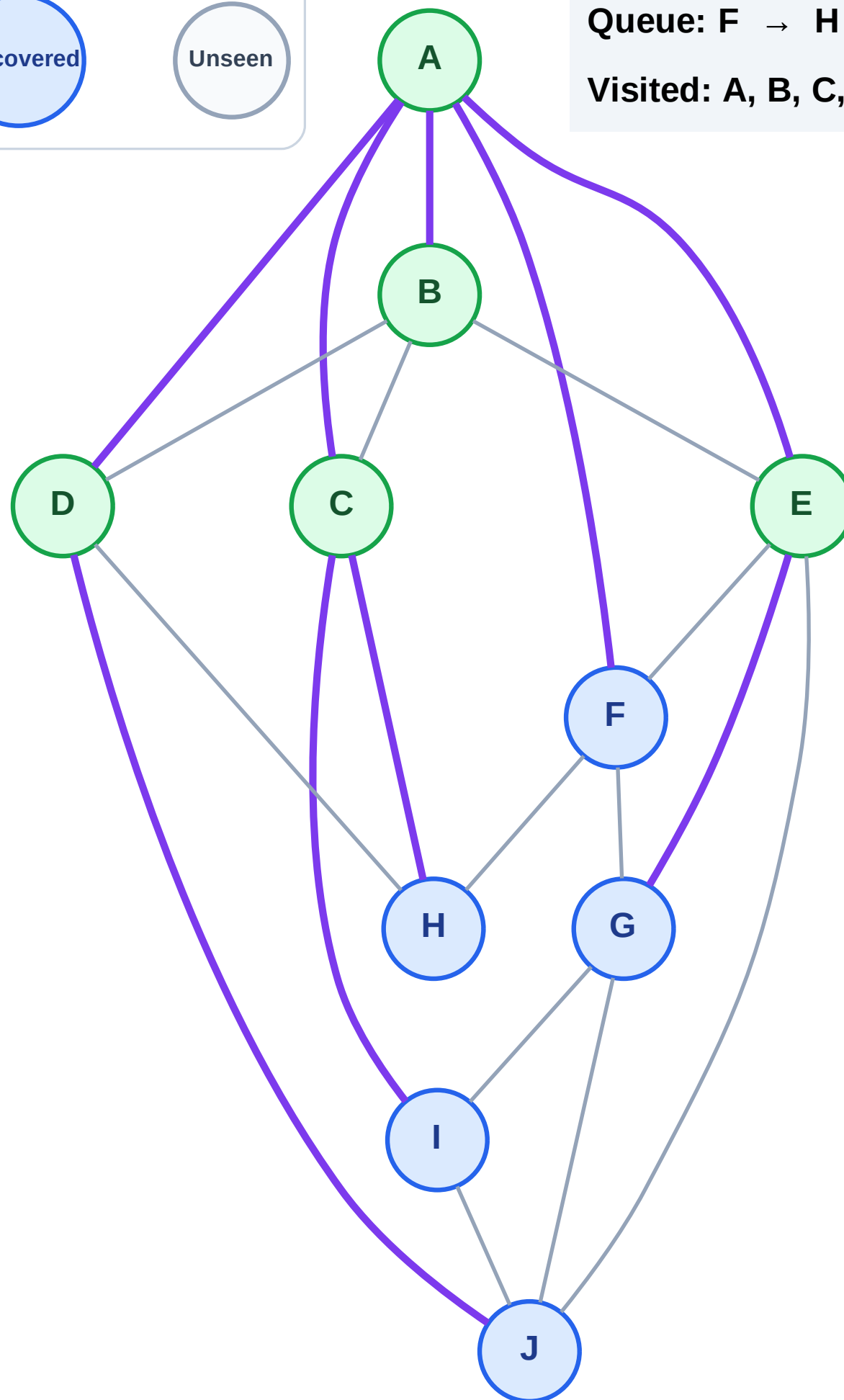
Step 11: Discover G from E

Legend



Queue: F → H → I → J → G

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node F

Legend

Complete

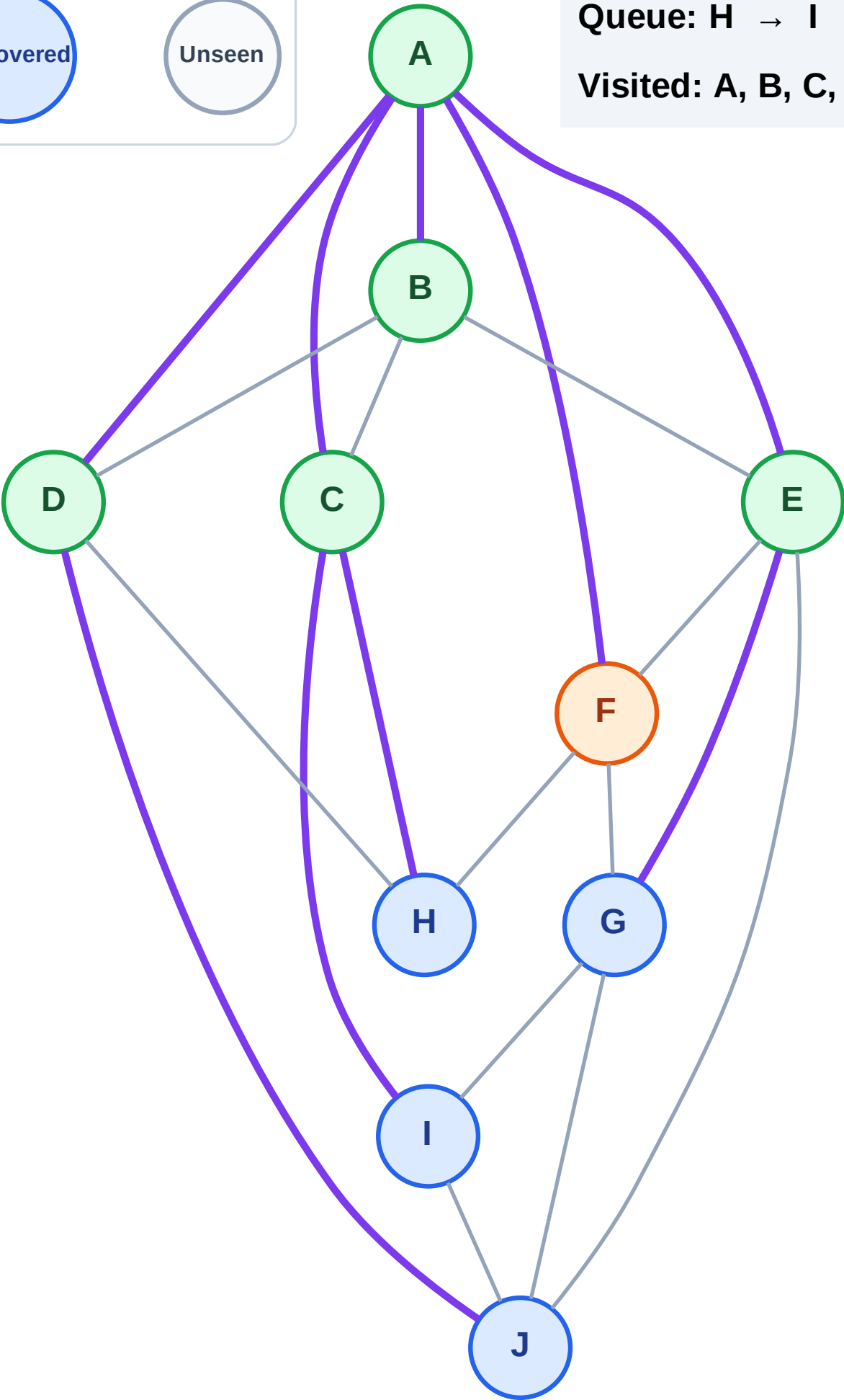
Processing

Discovered

Unseen

Queue: H → I → J → G

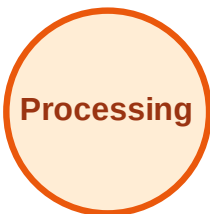
Visited: A, B, C, D, E



BFS Traversal

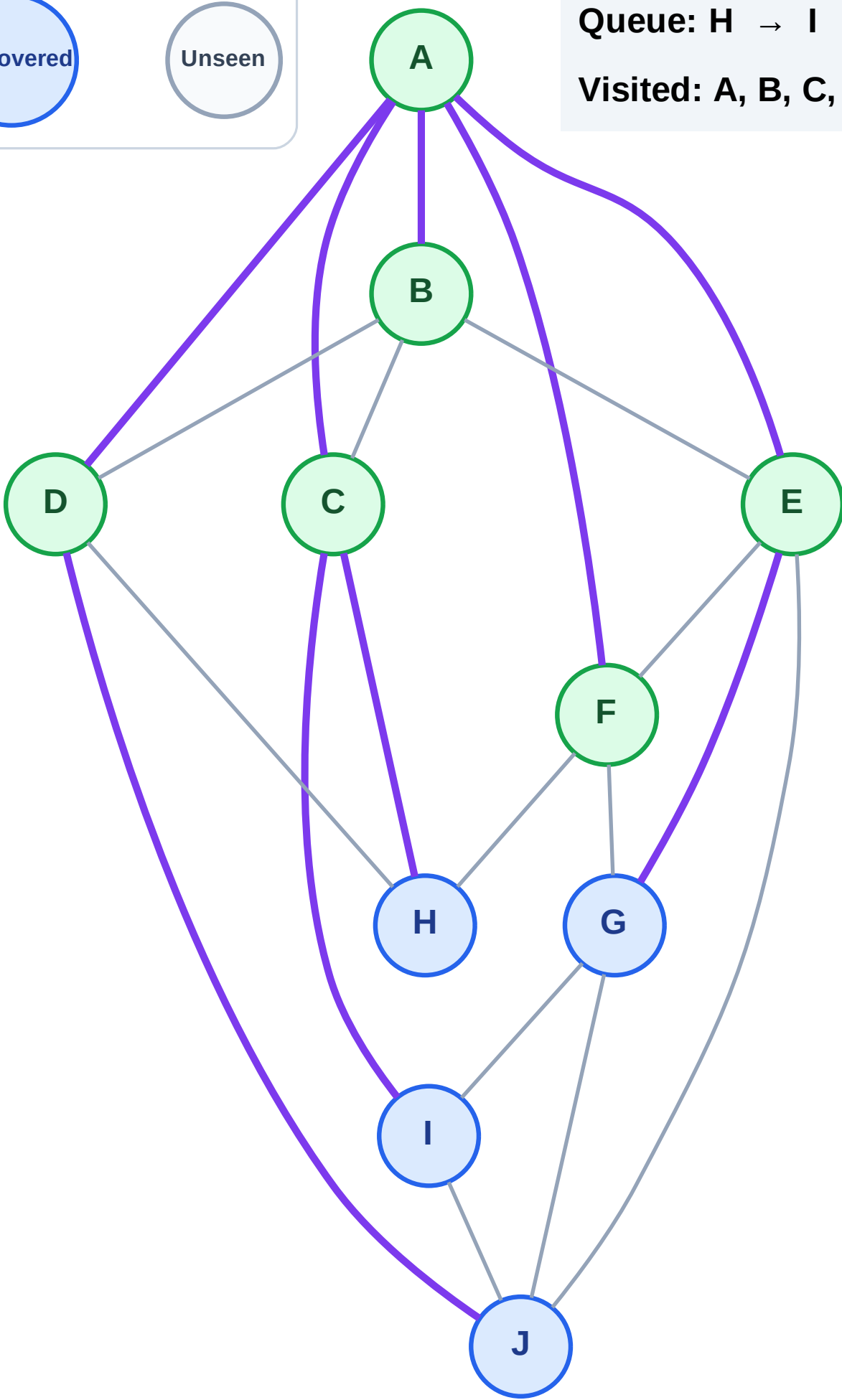
Step 13: No new neighbors from F

Legend



Queue: H → I → J → G

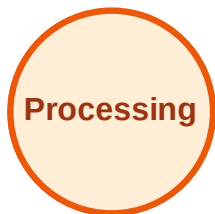
Visited: A, B, C, D, E, F



BFS Traversal

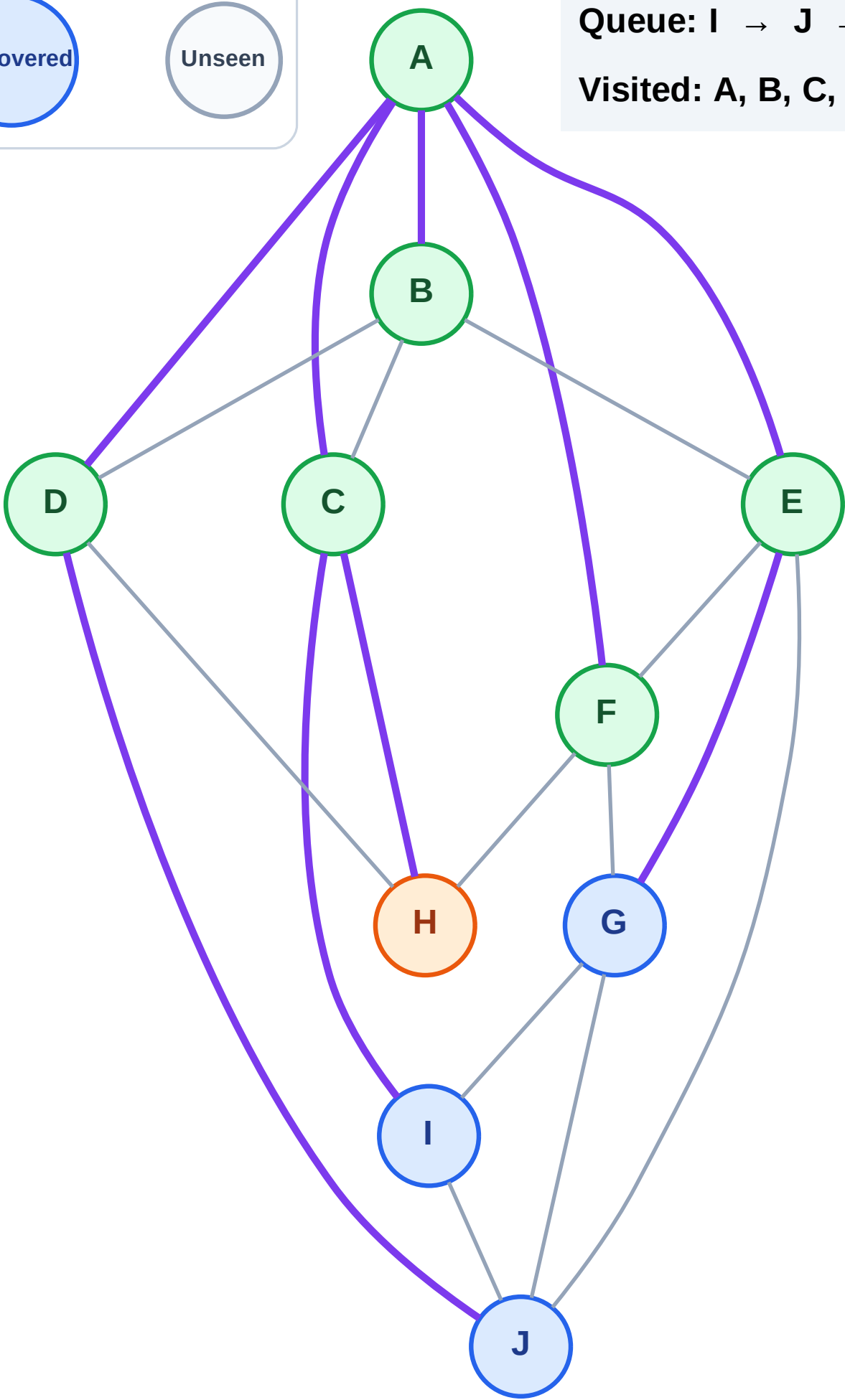
Step 14: Process node H

Legend



Queue: I → J → G

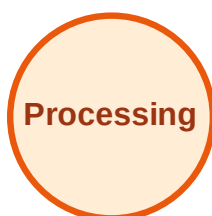
Visited: A, B, C, D, E, F



BFS Traversal

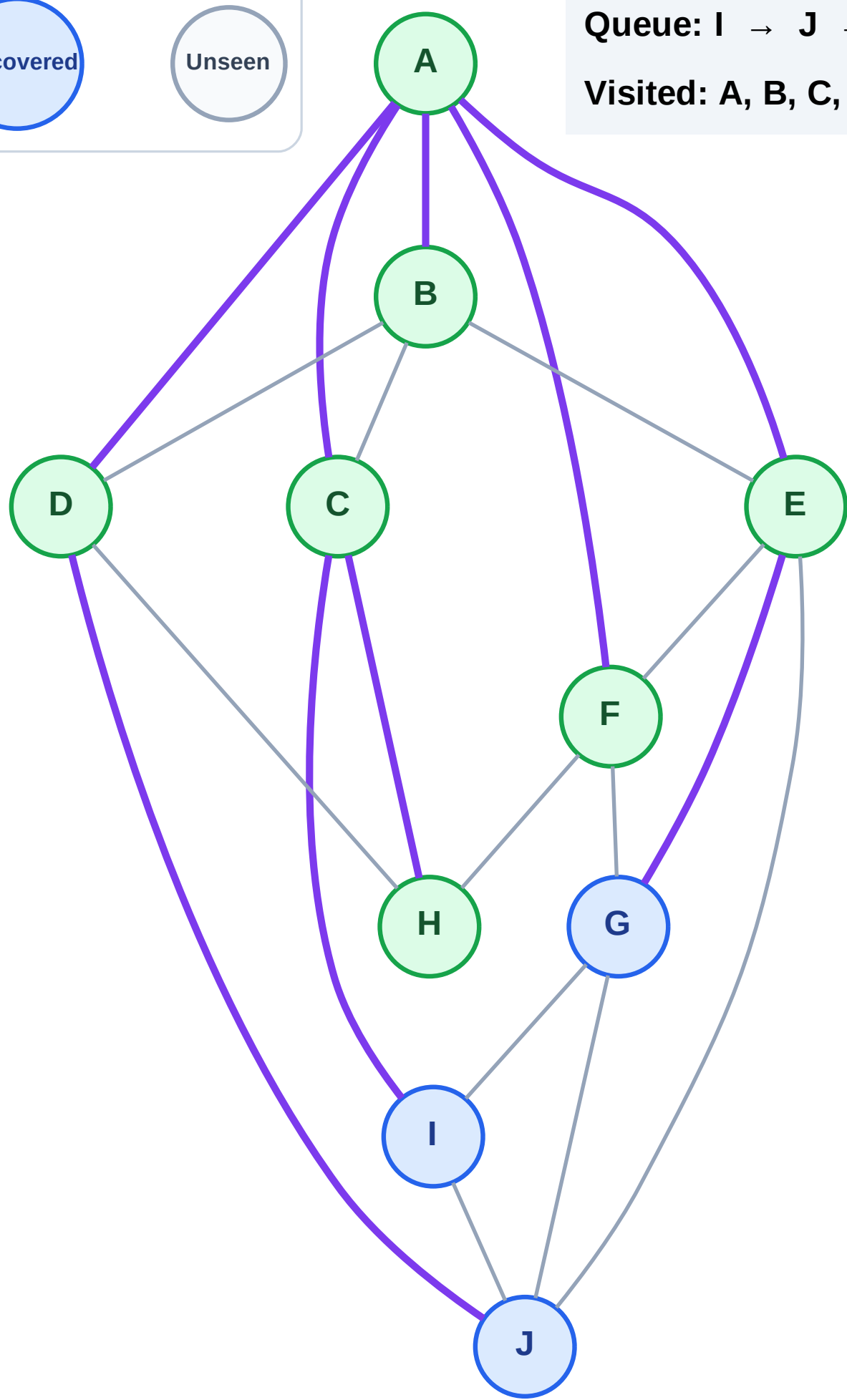
Step 15: No new neighbors from H

Legend



Queue: I → J → G

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node I

Legend

Complete

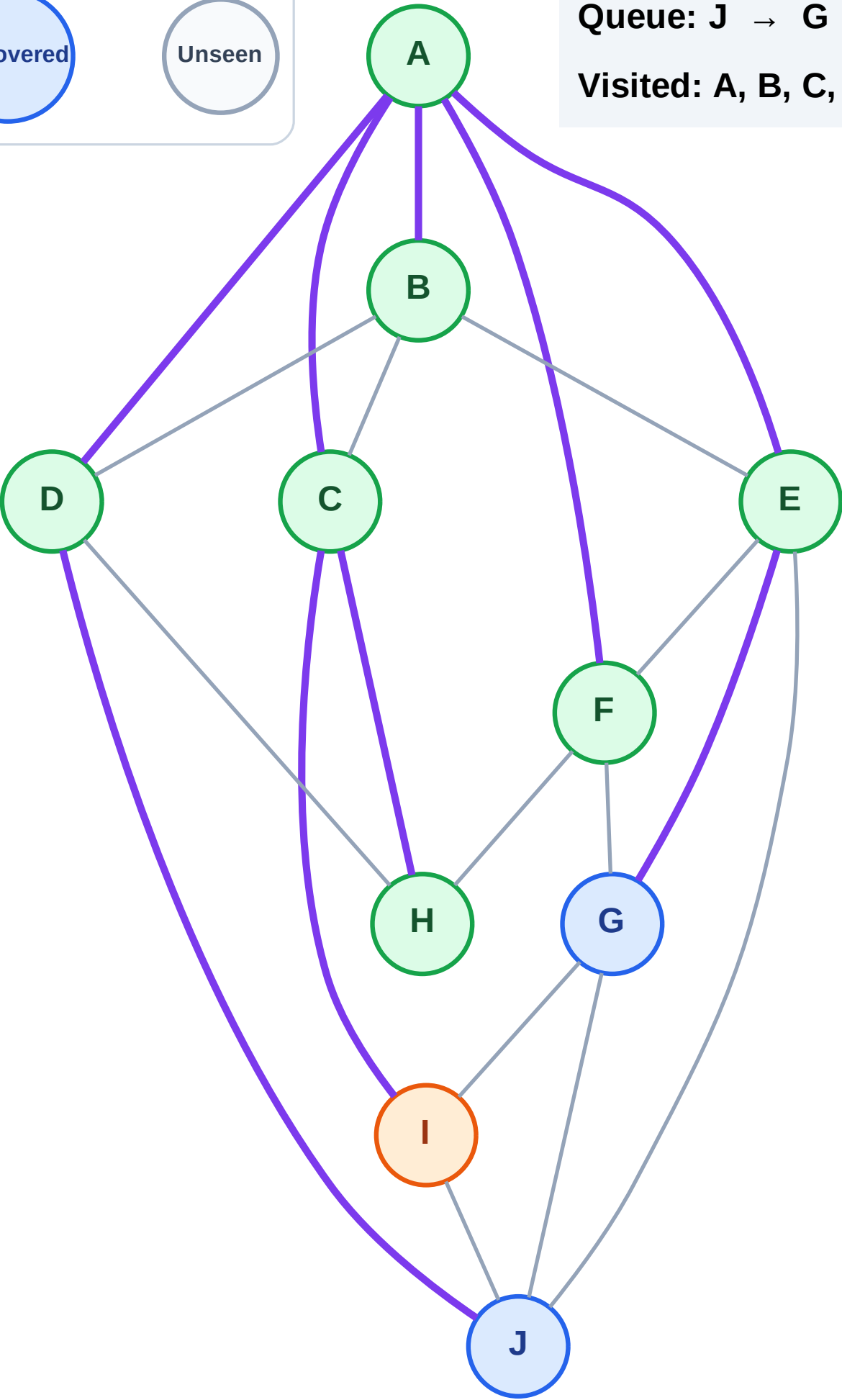
Processing

Discovered

Unseen

Queue: J → G

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

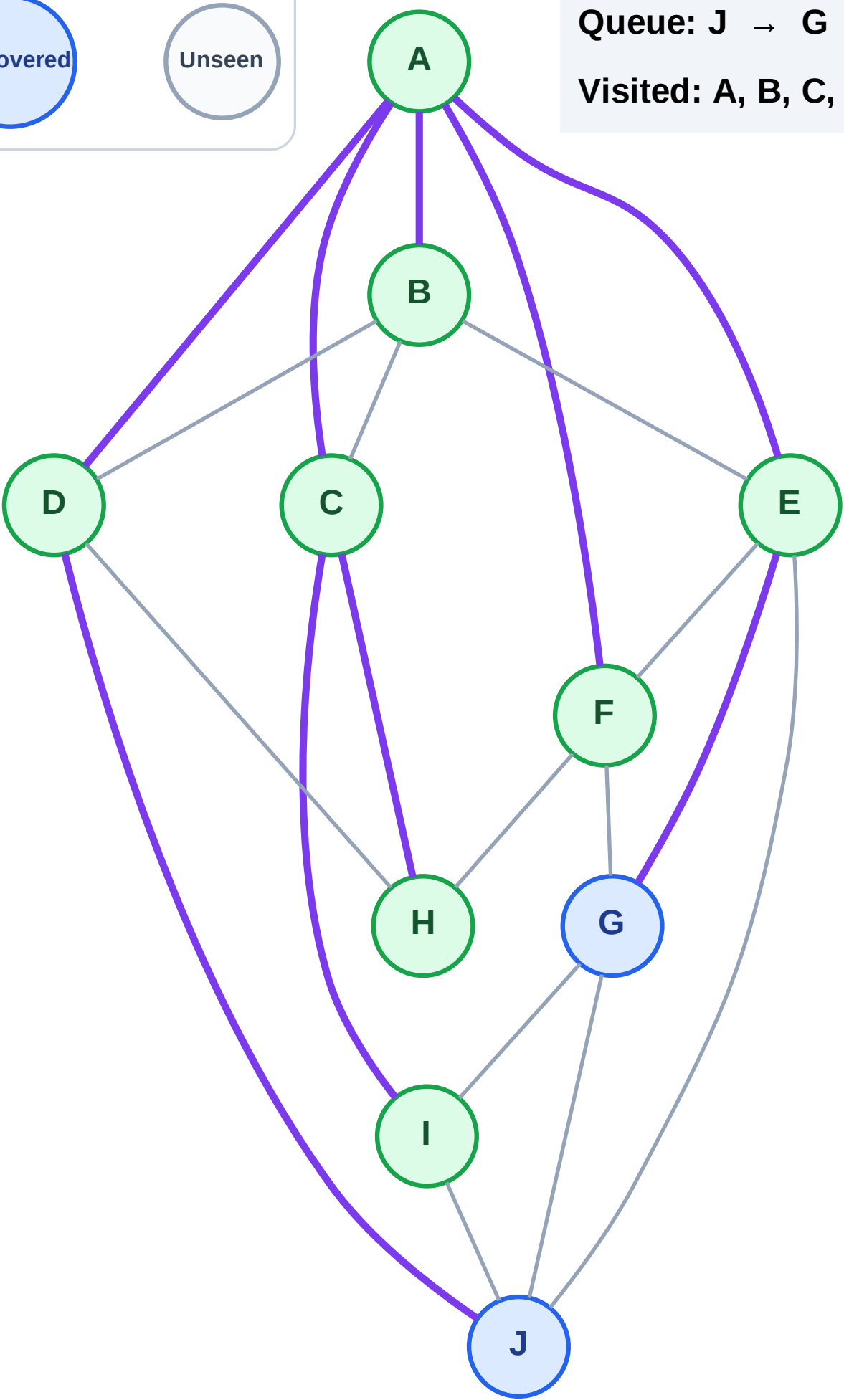
Processing

Discovered

Unseen

Queue: J → G

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 18: Process node J

Legend

Complete

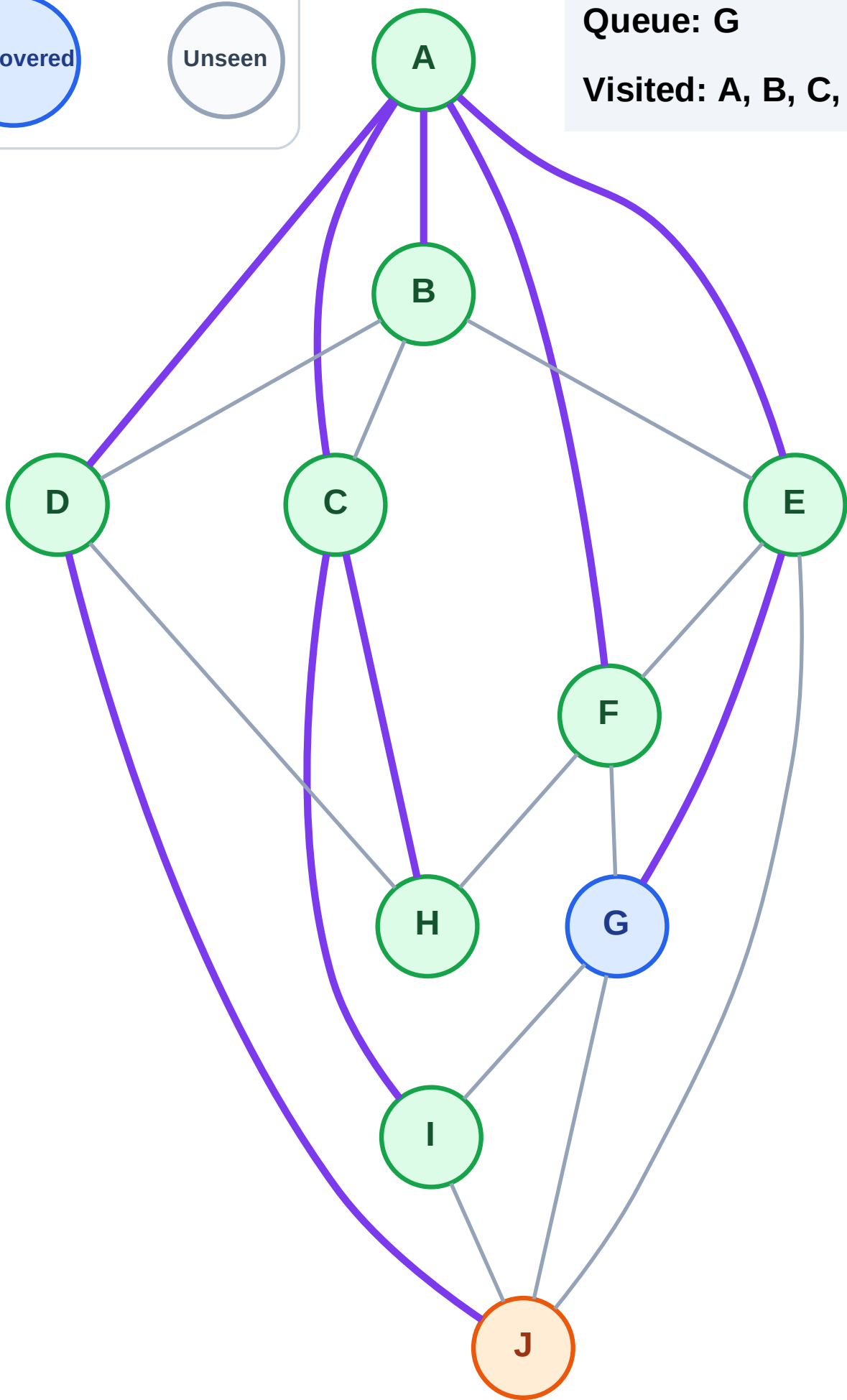
Processing

Discovered

Unseen

Queue: G

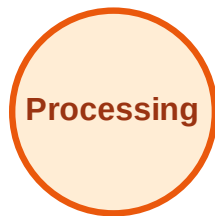
Visited: A, B, C, D, E, F, H, I



BFS Traversal

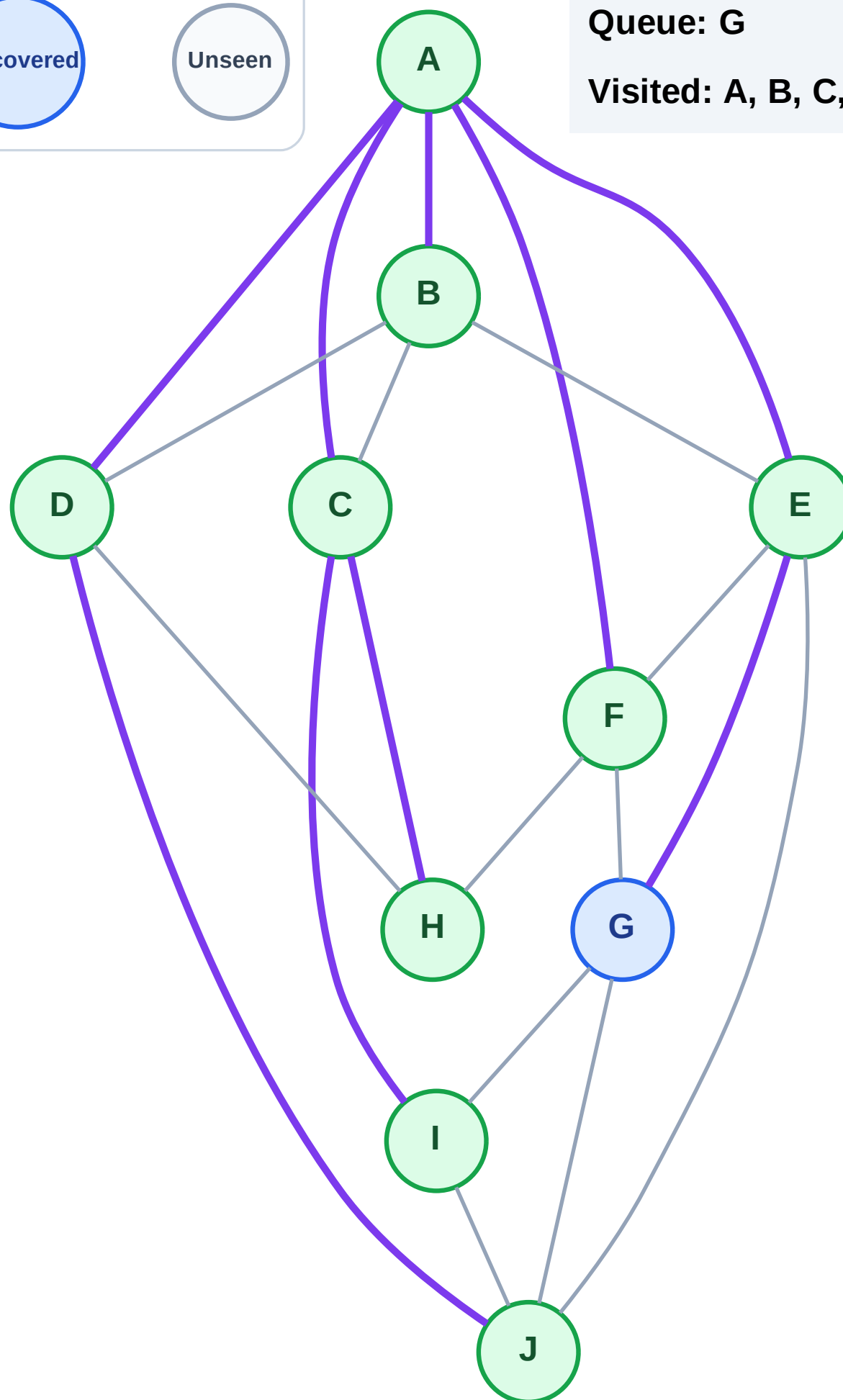
Step 19: No new neighbors from J

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 20: Process node G

Legend

Complete

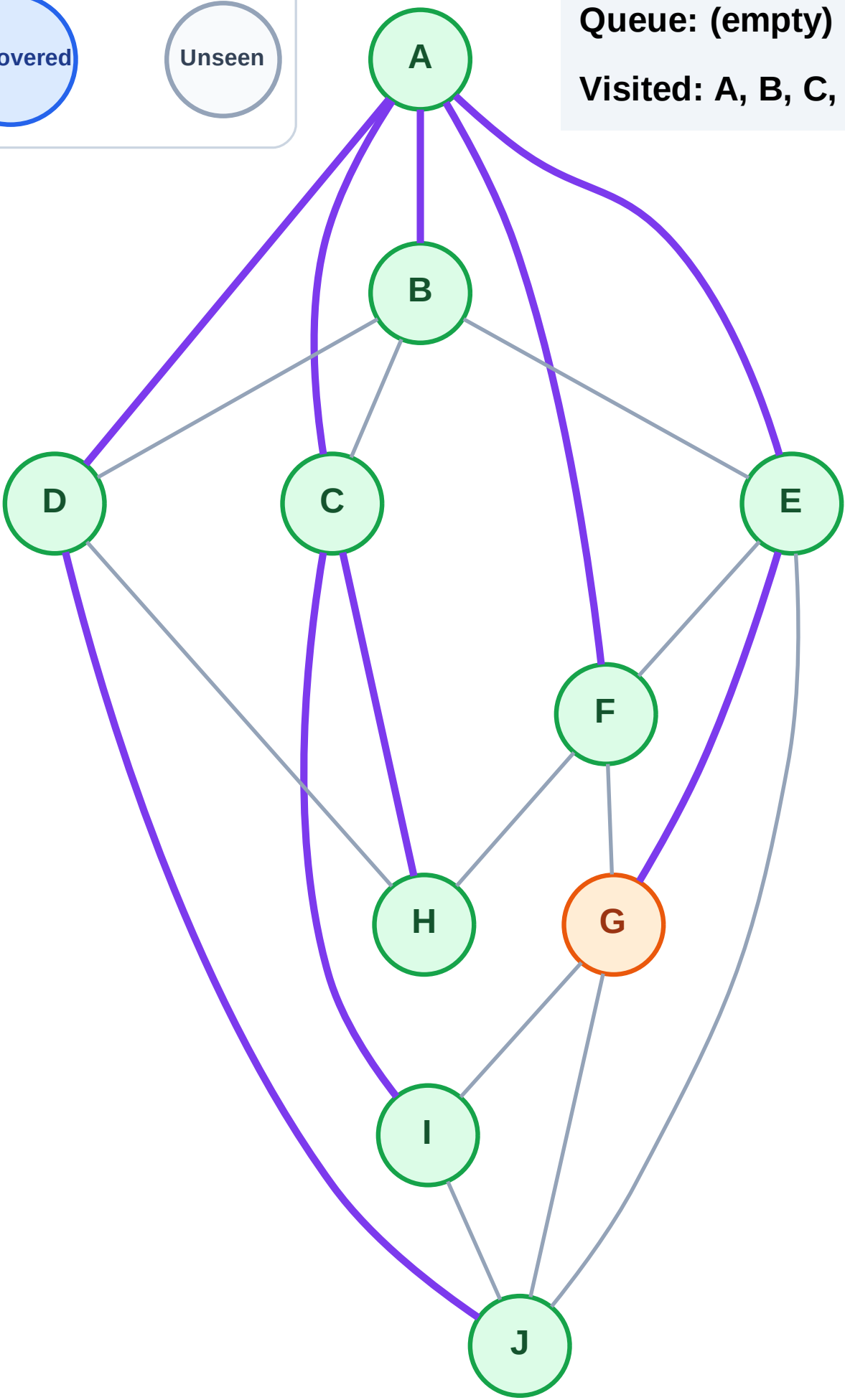
Processing

Discovered

Unseen

Queue: (empty)

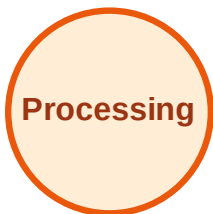
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

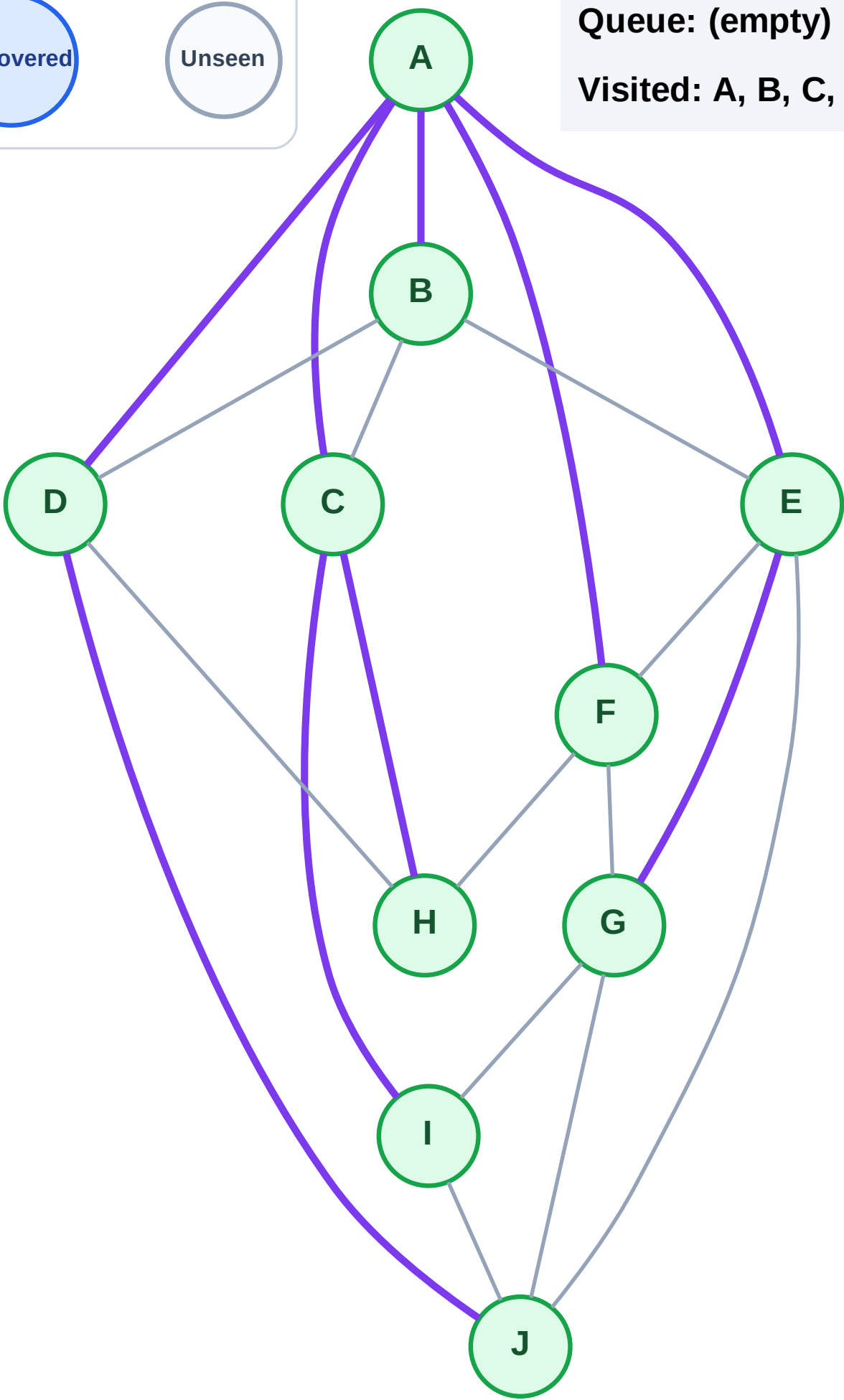
Step 21: No new neighbors from G

Legend



Queue: (empty)

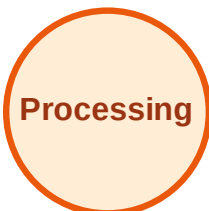
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

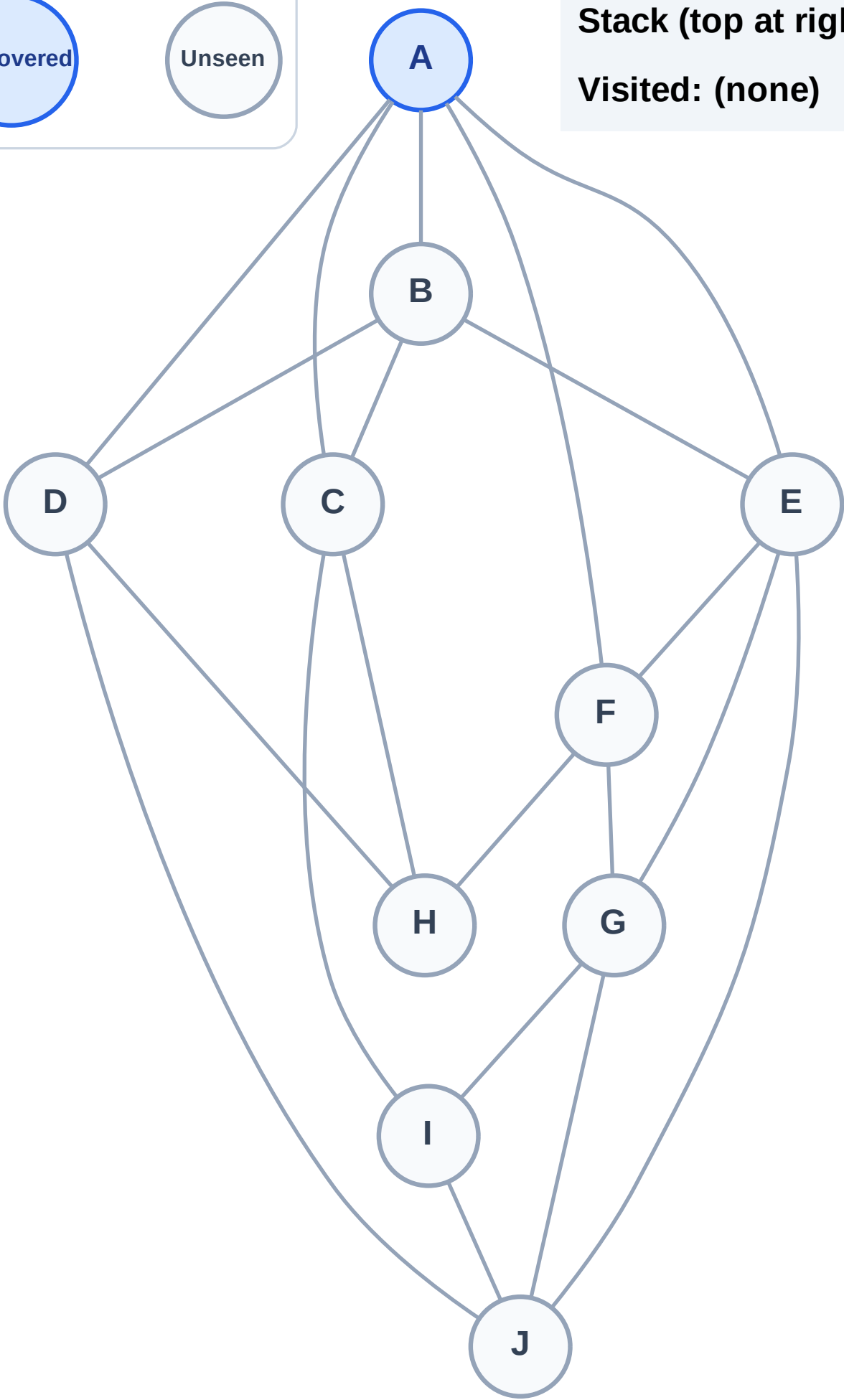
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

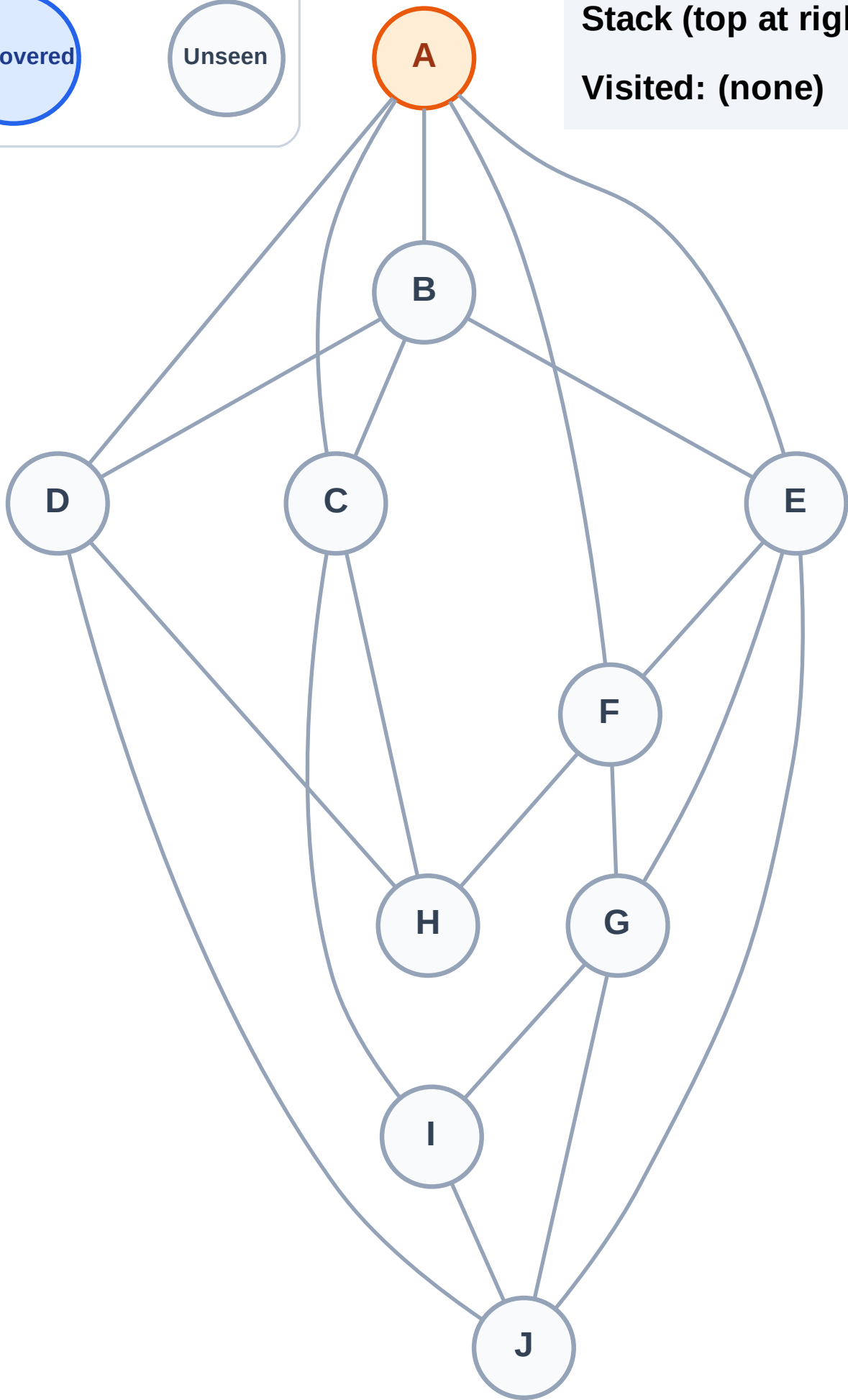
Complete

Processing

Discovered

Unseen

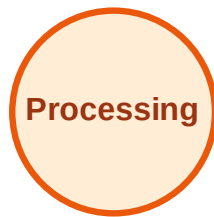
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

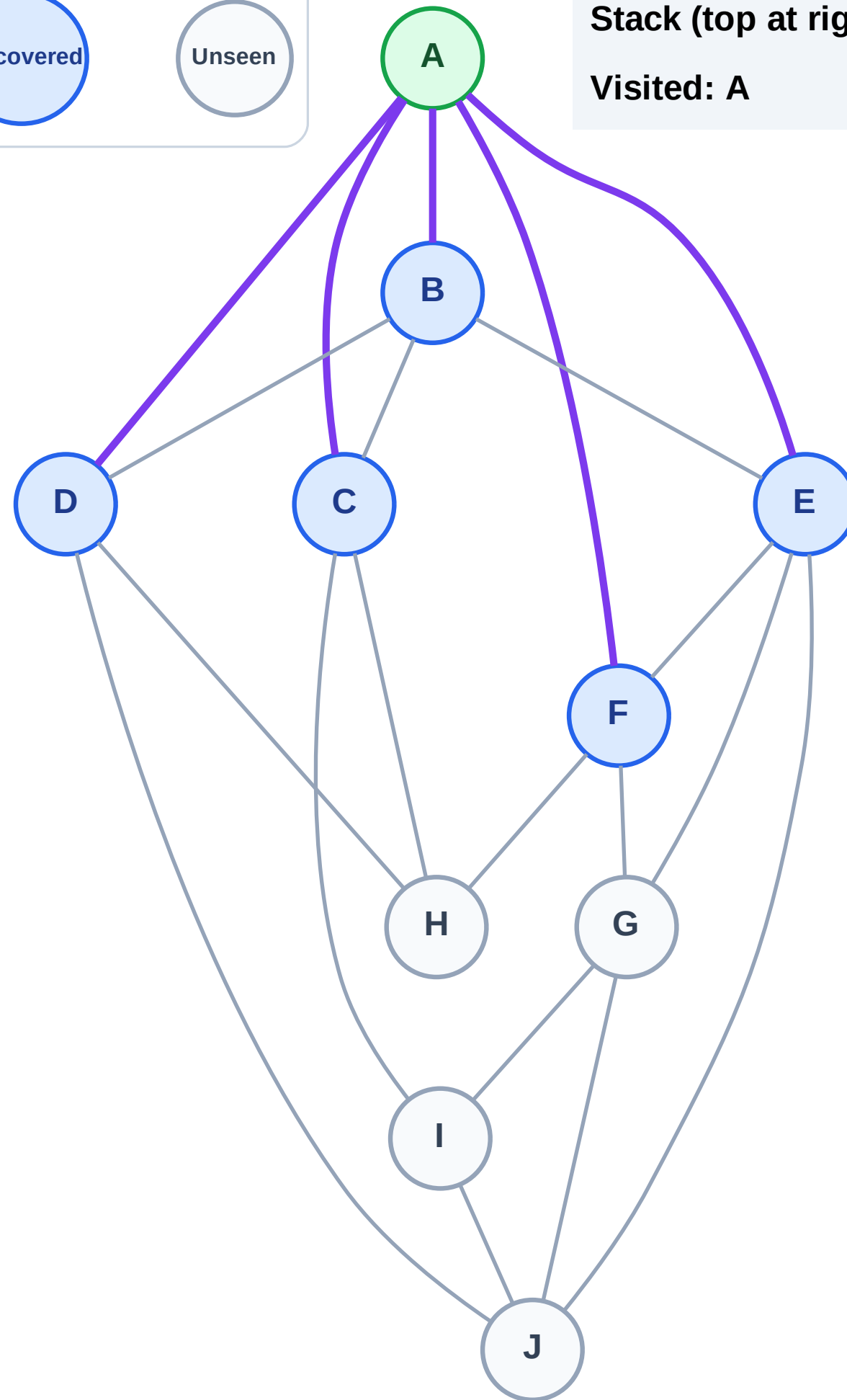
Step 3: Discover F, E, D, C, B from A

Legend



Stack (top at right): F | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

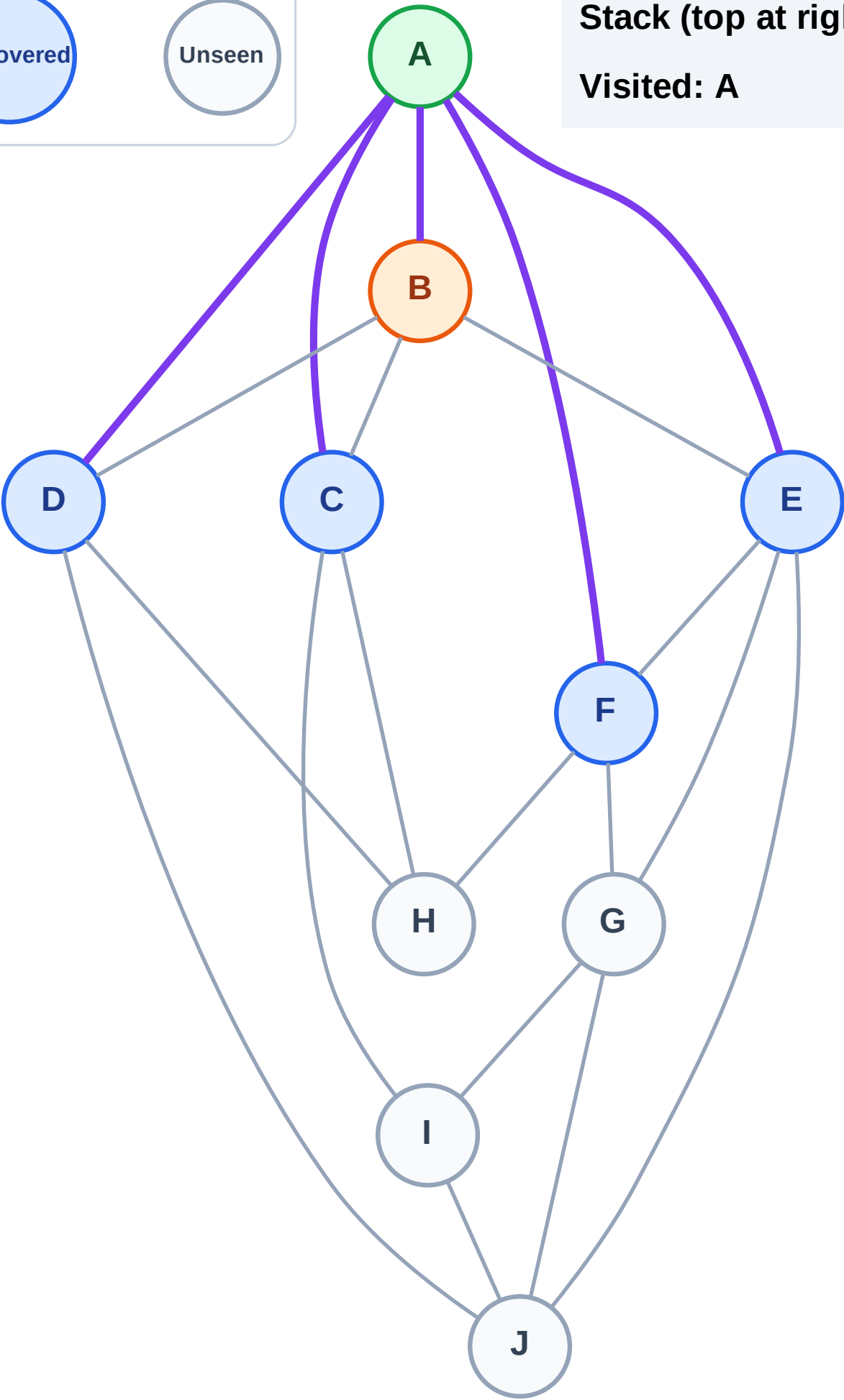
Processing

Discovered

Unseen

Stack (top at right): F | E | D | C

Visited: A



DFS Traversal

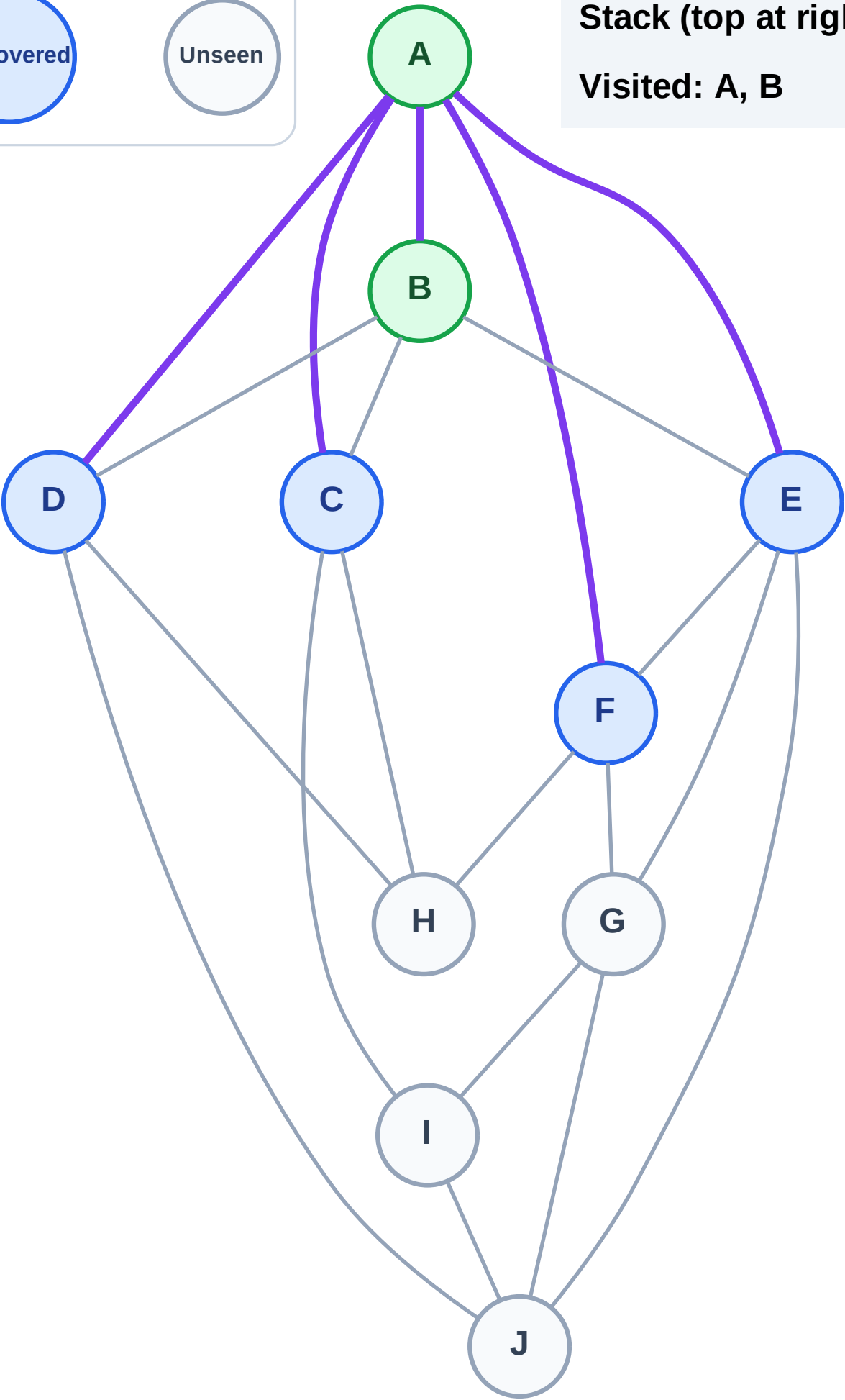
Step 5: No new neighbors from B

Legend



Stack (top at right): F | E | D | C

Visited: A, B



DFS Traversal

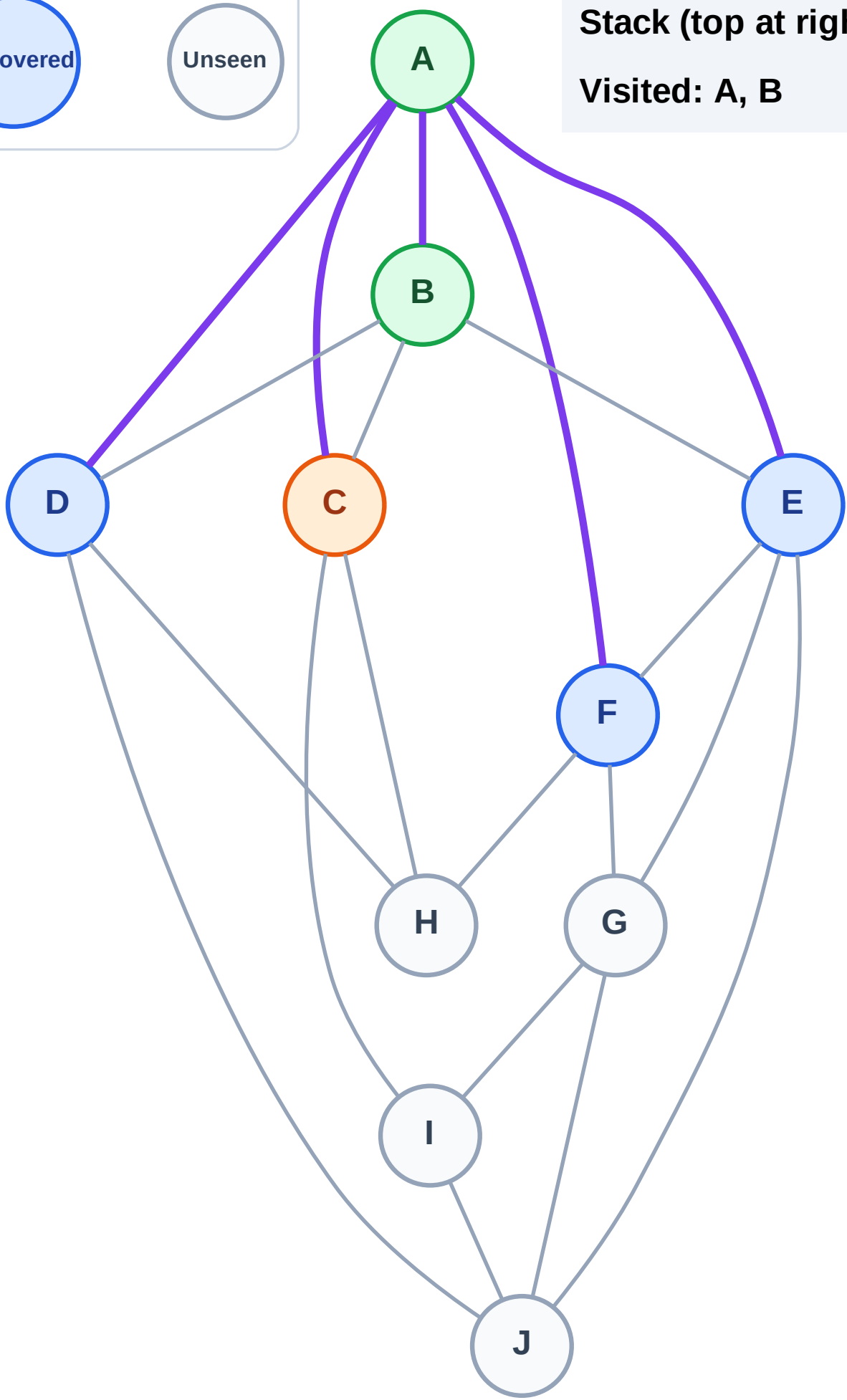
Step 6: Process node C

Legend



Stack (top at right): F | E | D

Visited: A, B



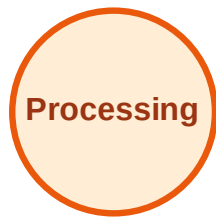
DFS Traversal

Step 7: Discover I, H from C

Legend



Complete



Processing



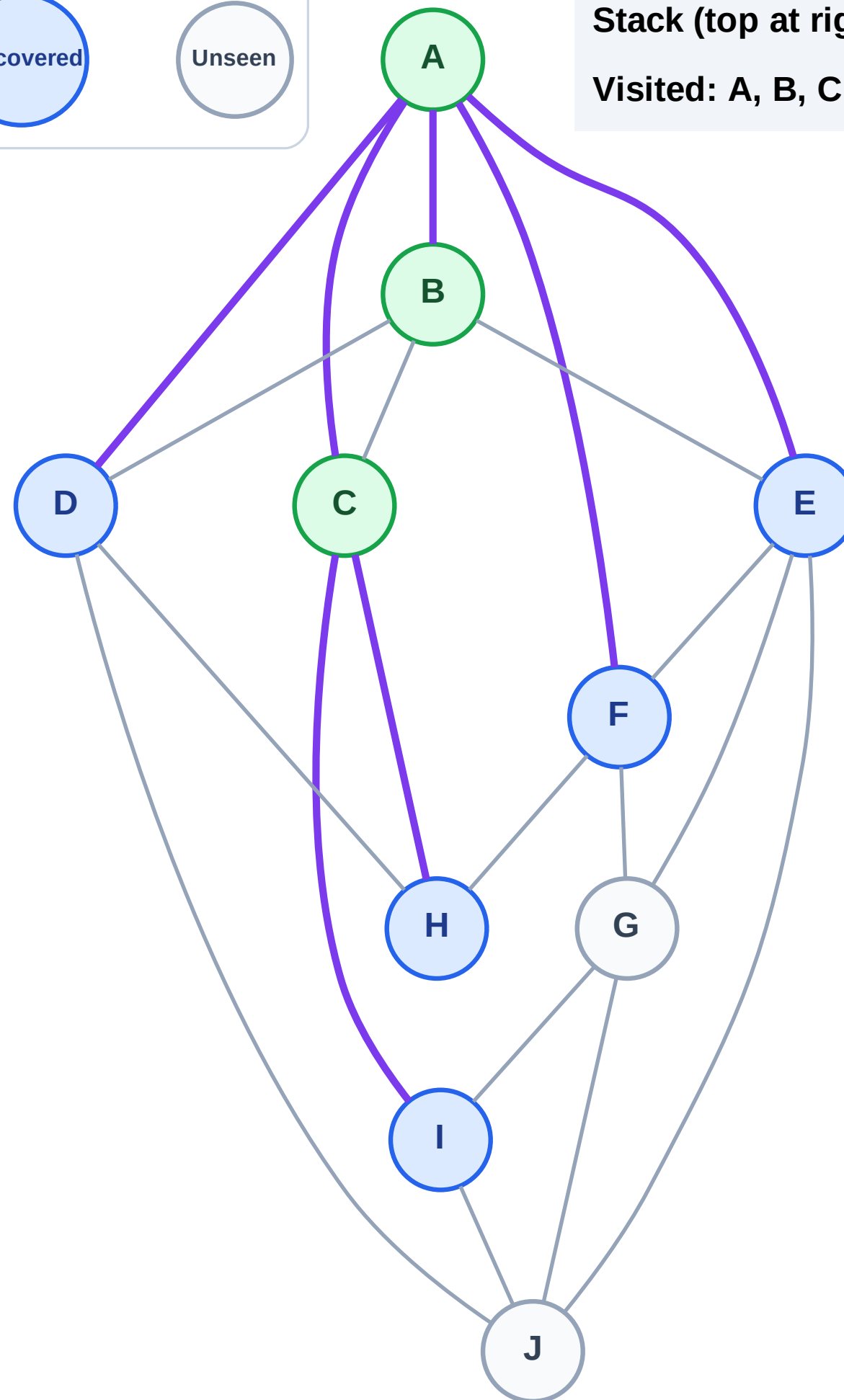
Discovered



Unseen

Stack (top at right): F | E | D | I | H

Visited: A, B, C



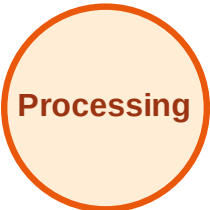
DFS Traversal

Step 8: Process node H

Legend



Processing



Discovered

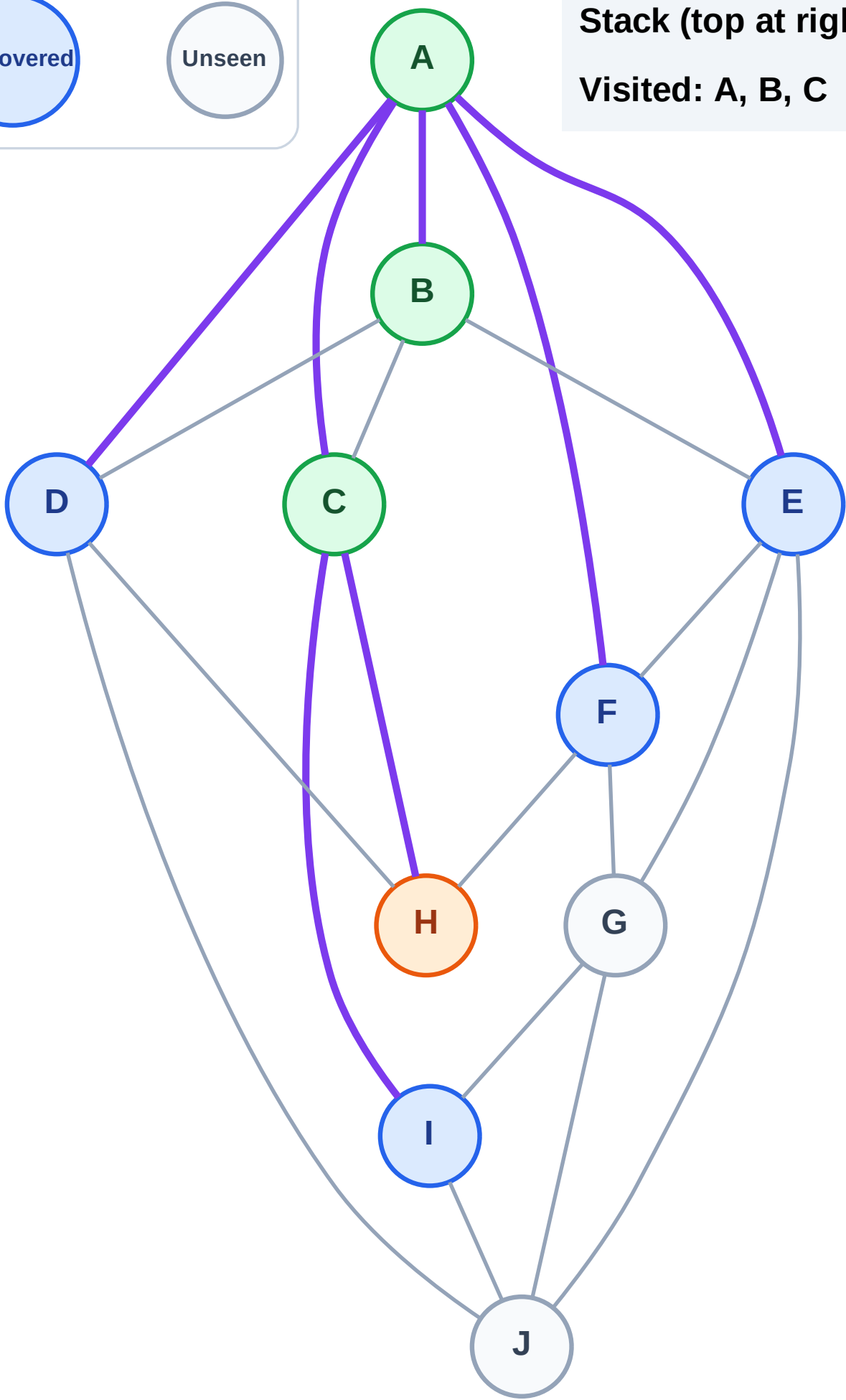


Unseen



Stack (top at right): F | E | D | I

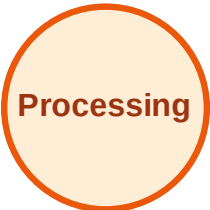
Visited: A, B, C



DFS Traversal

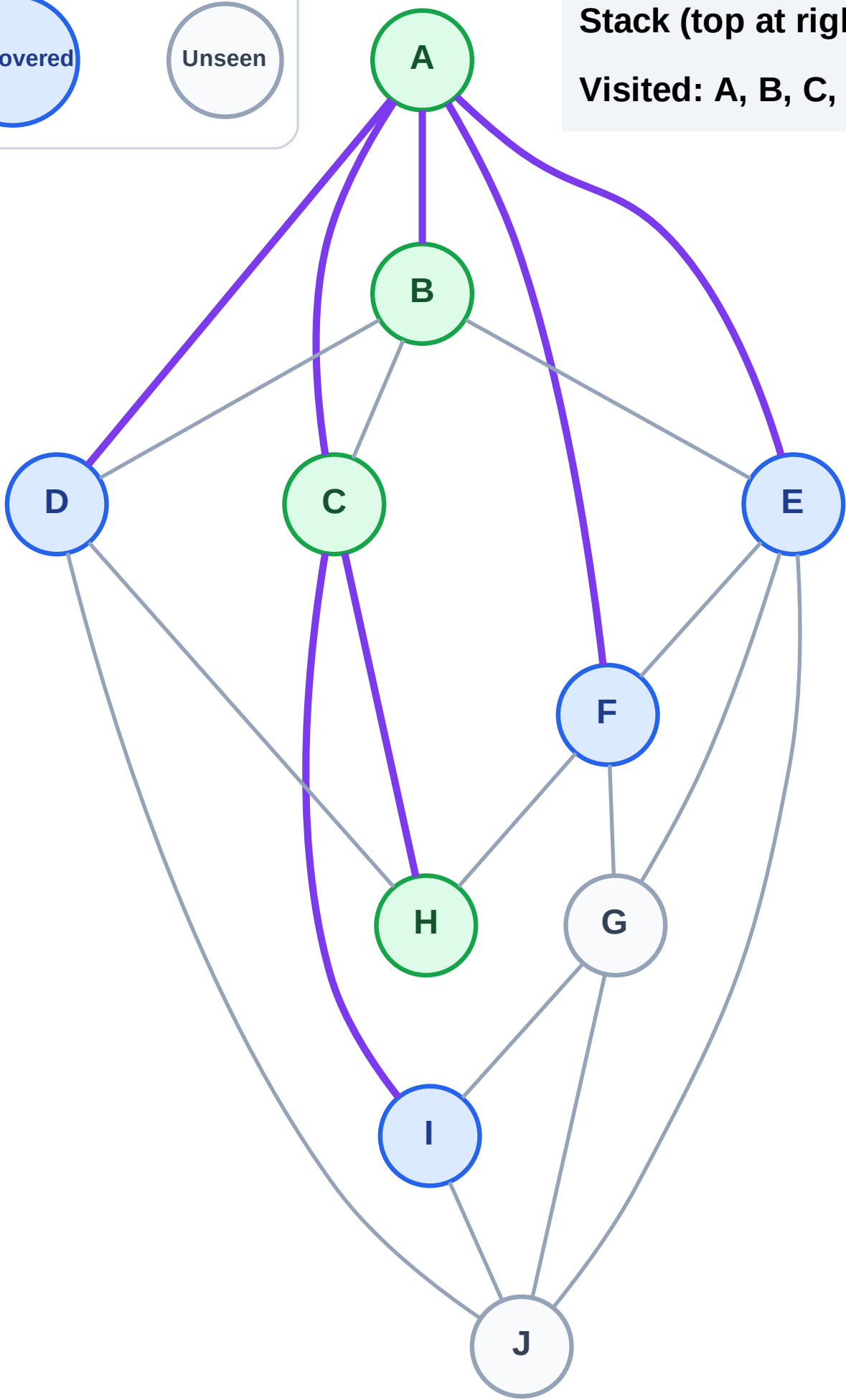
Step 9: No new neighbors from H

Legend



Stack (top at right): F | E | D | I

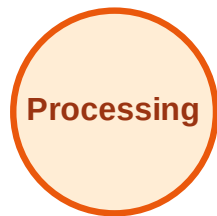
Visited: A, B, C, H



DFS Traversal

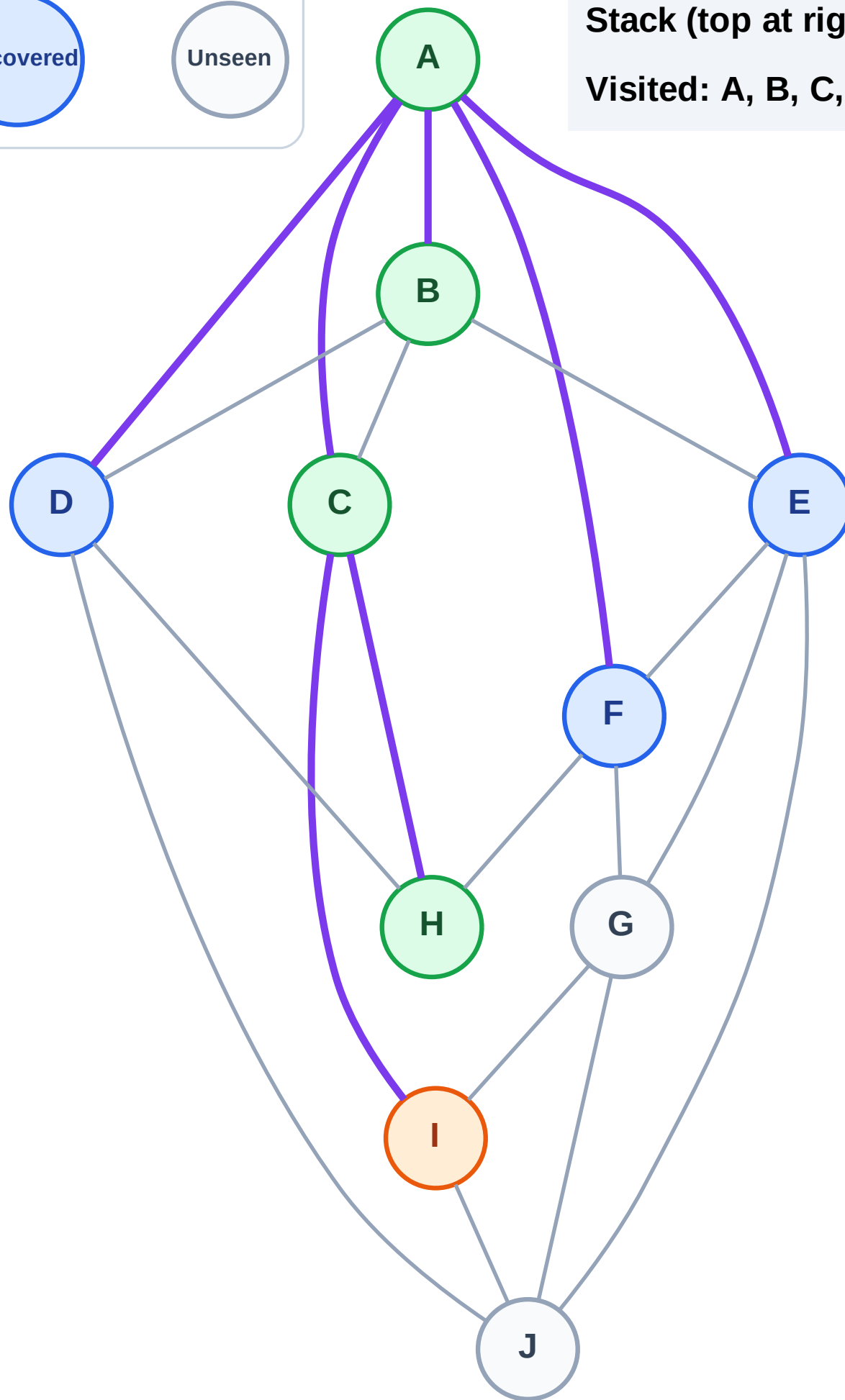
Step 10: Process node I

Legend



Stack (top at right): F | E | D

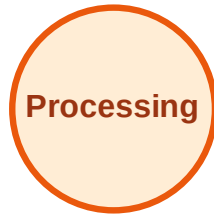
Visited: A, B, C, H



DFS Traversal

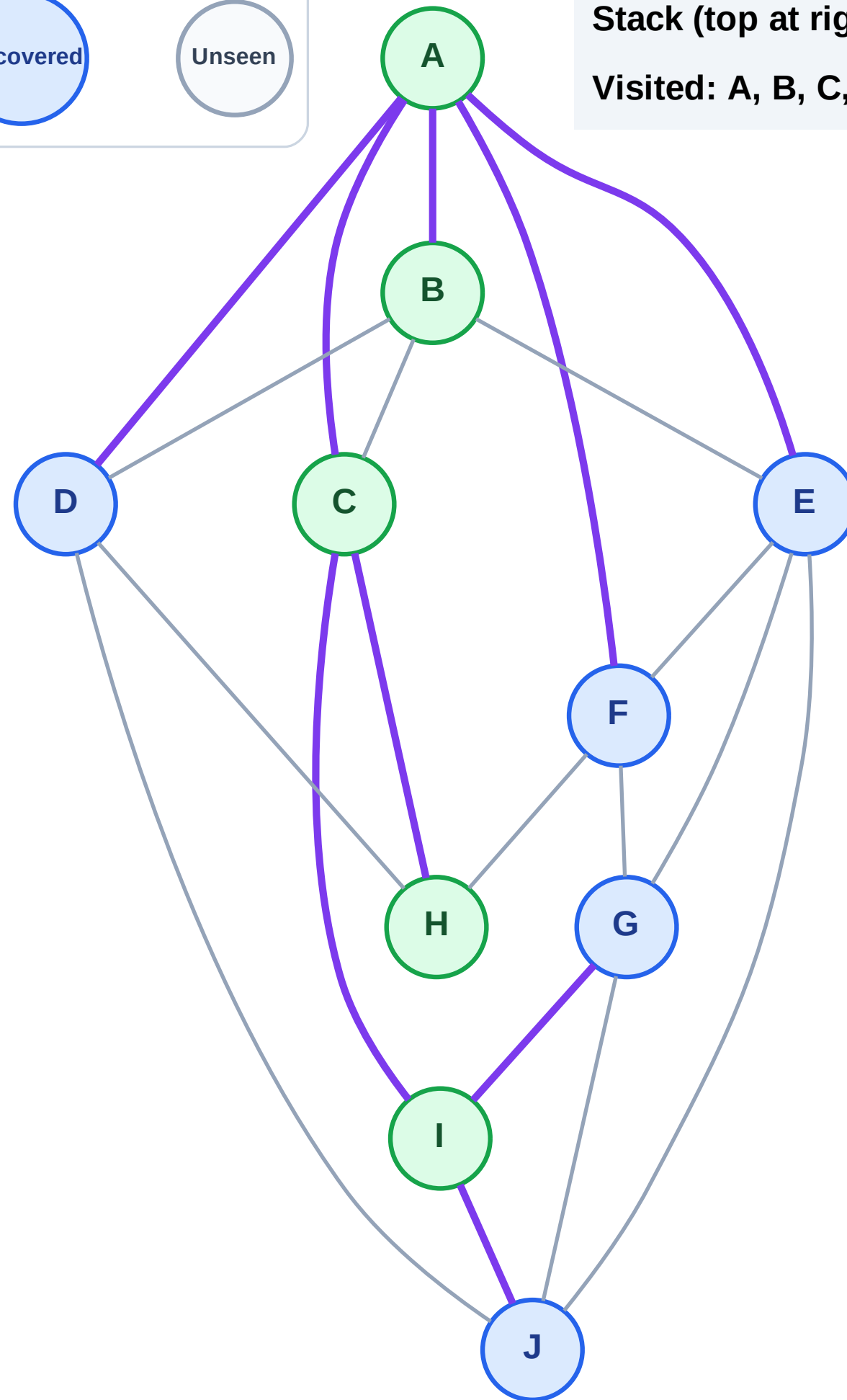
Step 11: Discover J, G from I

Legend



Stack (top at right): F | E | D | J | G

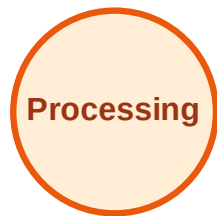
Visited: A, B, C, H, I



DFS Traversal

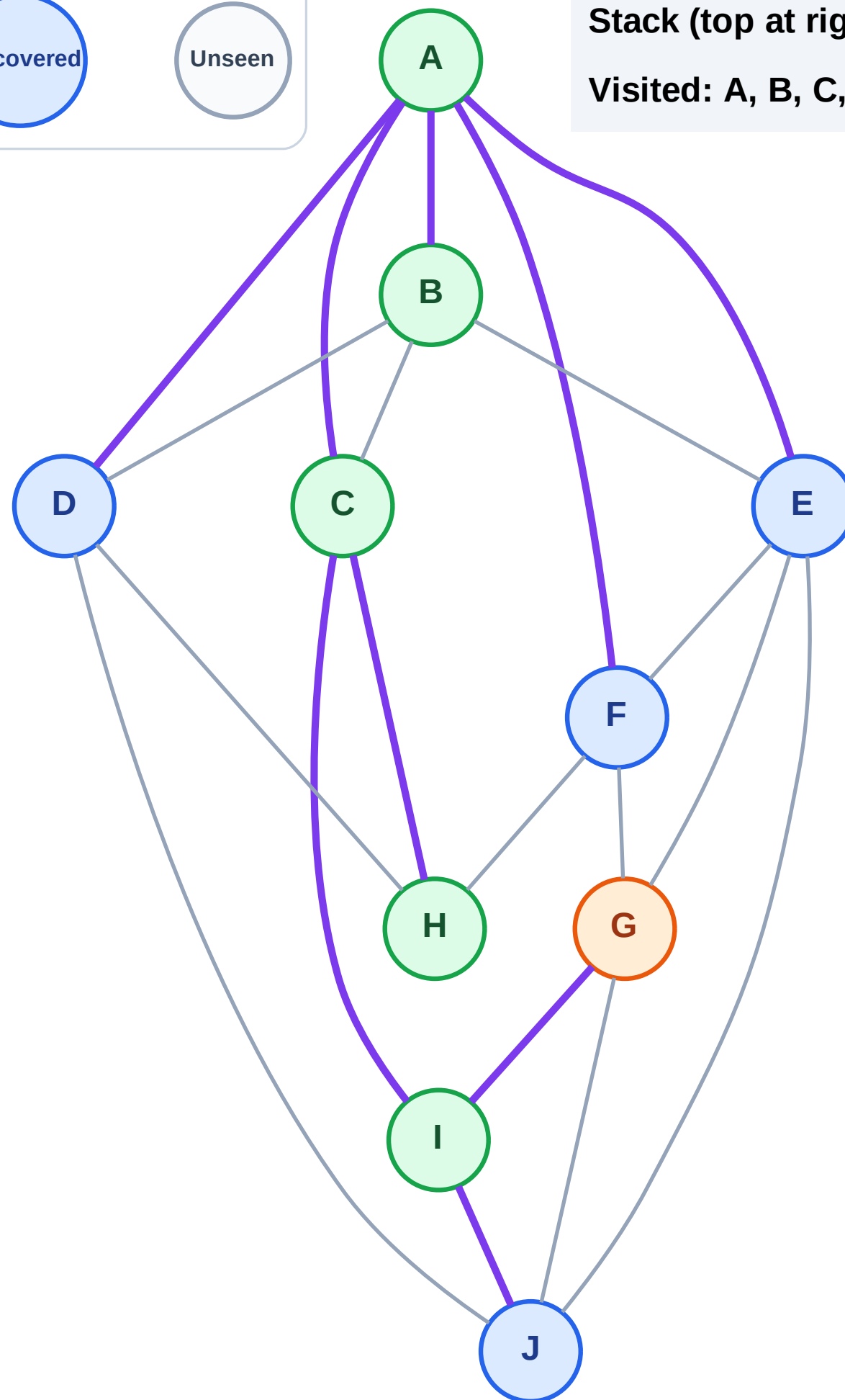
Step 12: Process node G

Legend



Stack (top at right): F | E | D | J

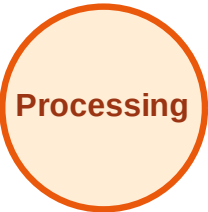
Visited: A, B, C, H, I



DFS Traversal

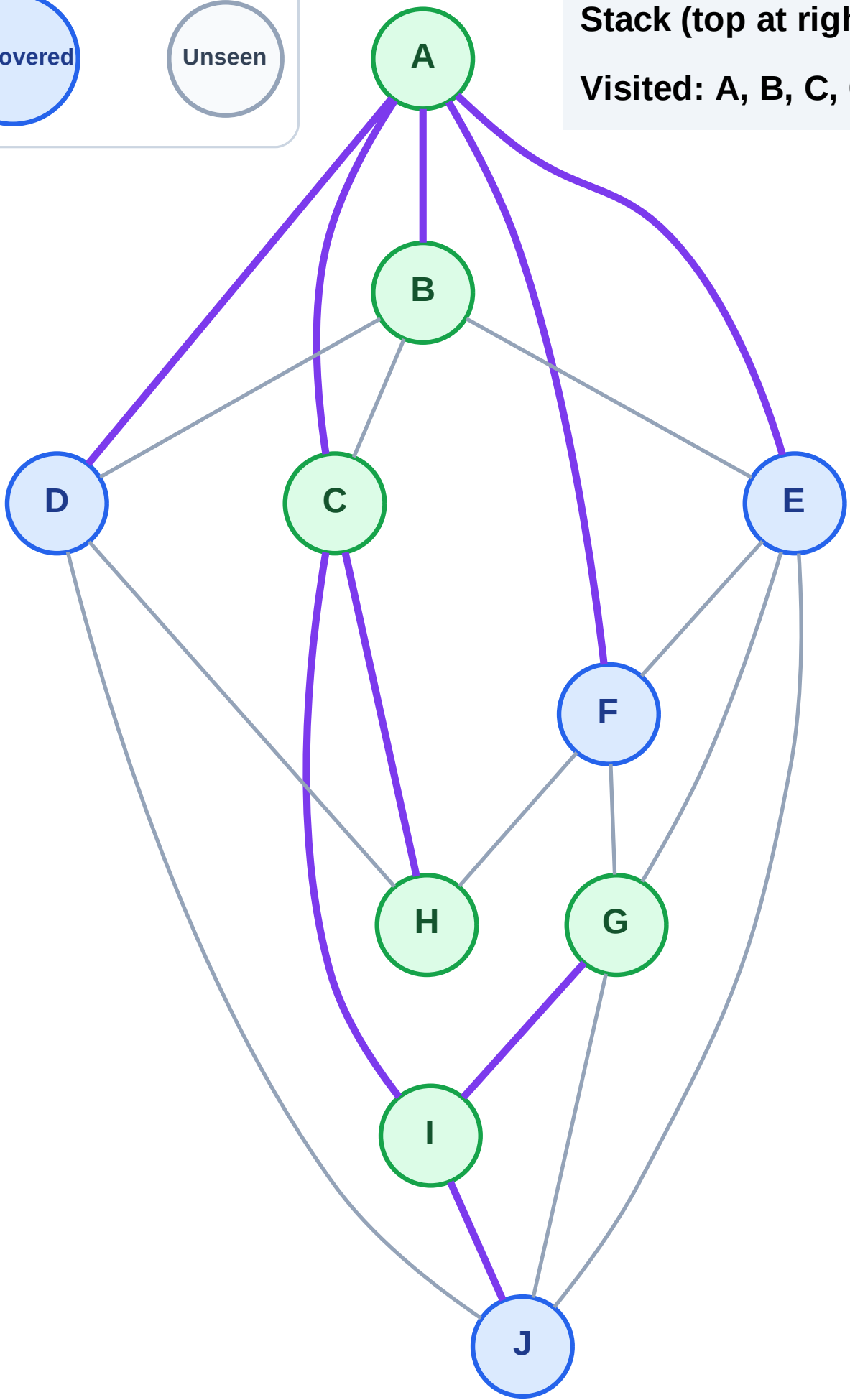
Step 13: No new neighbors from G

Legend



Stack (top at right): F | E | D | J

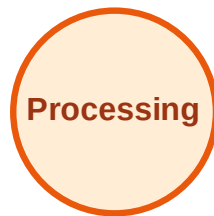
Visited: A, B, C, G, H, I



DFS Traversal

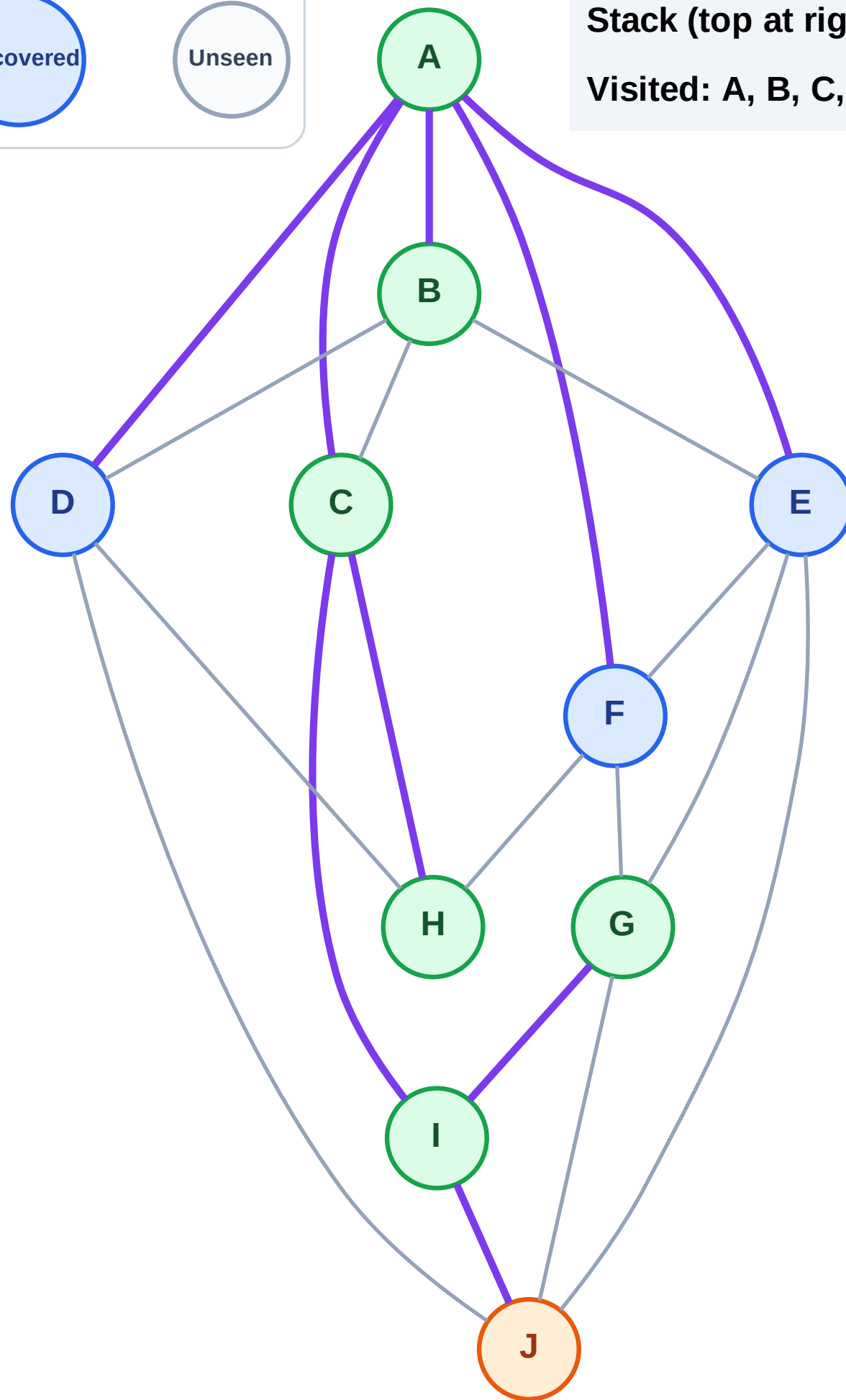
Step 14: Process node J

Legend



Stack (top at right): F | E | D

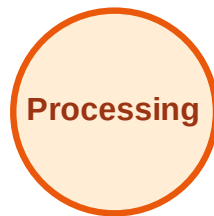
Visited: A, B, C, G, H, I



DFS Traversal

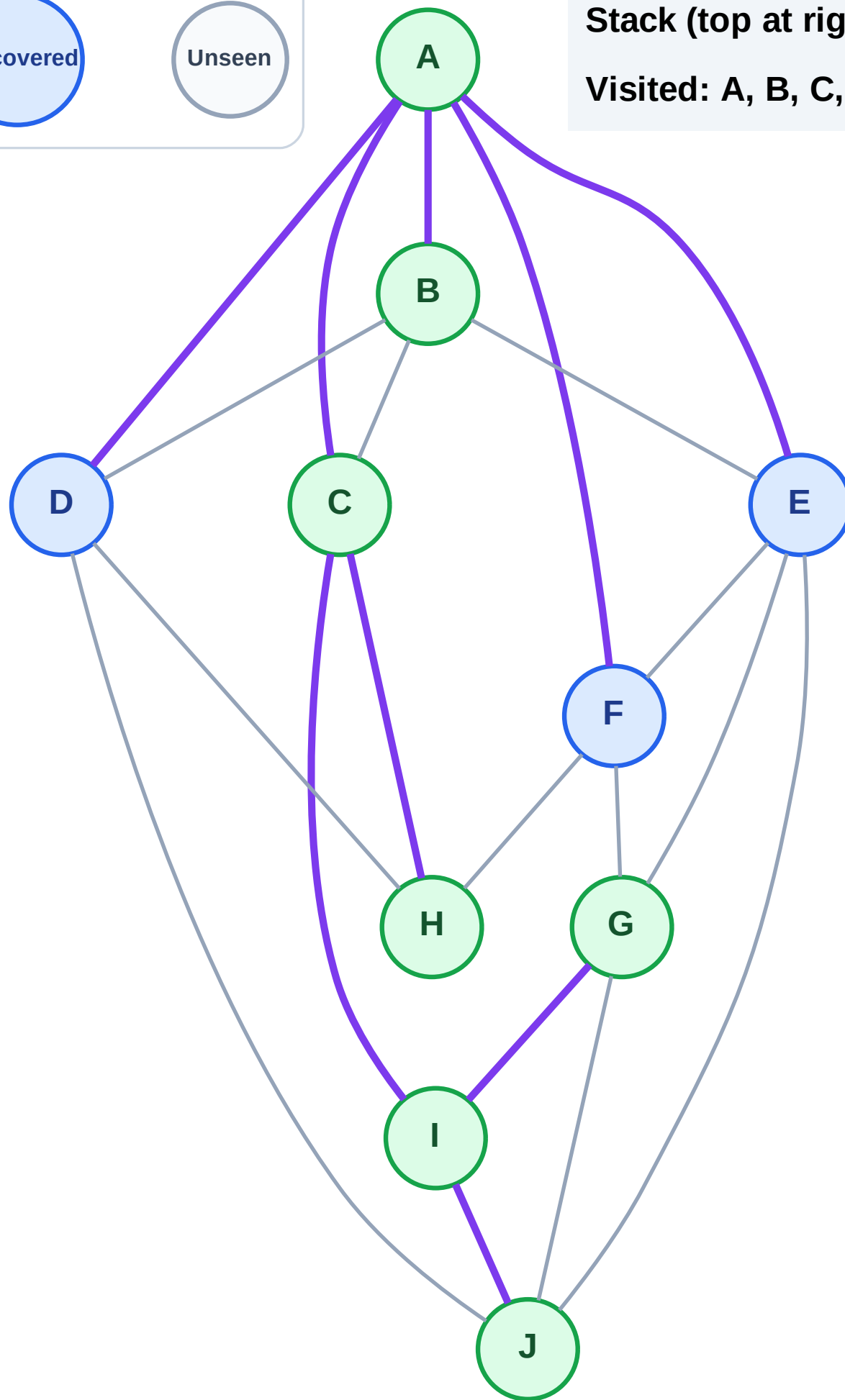
Step 15: No new neighbors from J

Legend



Stack (top at right): F | E | D

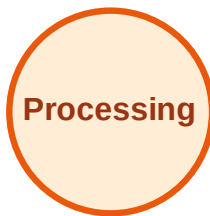
Visited: A, B, C, G, H, I, J



DFS Traversal

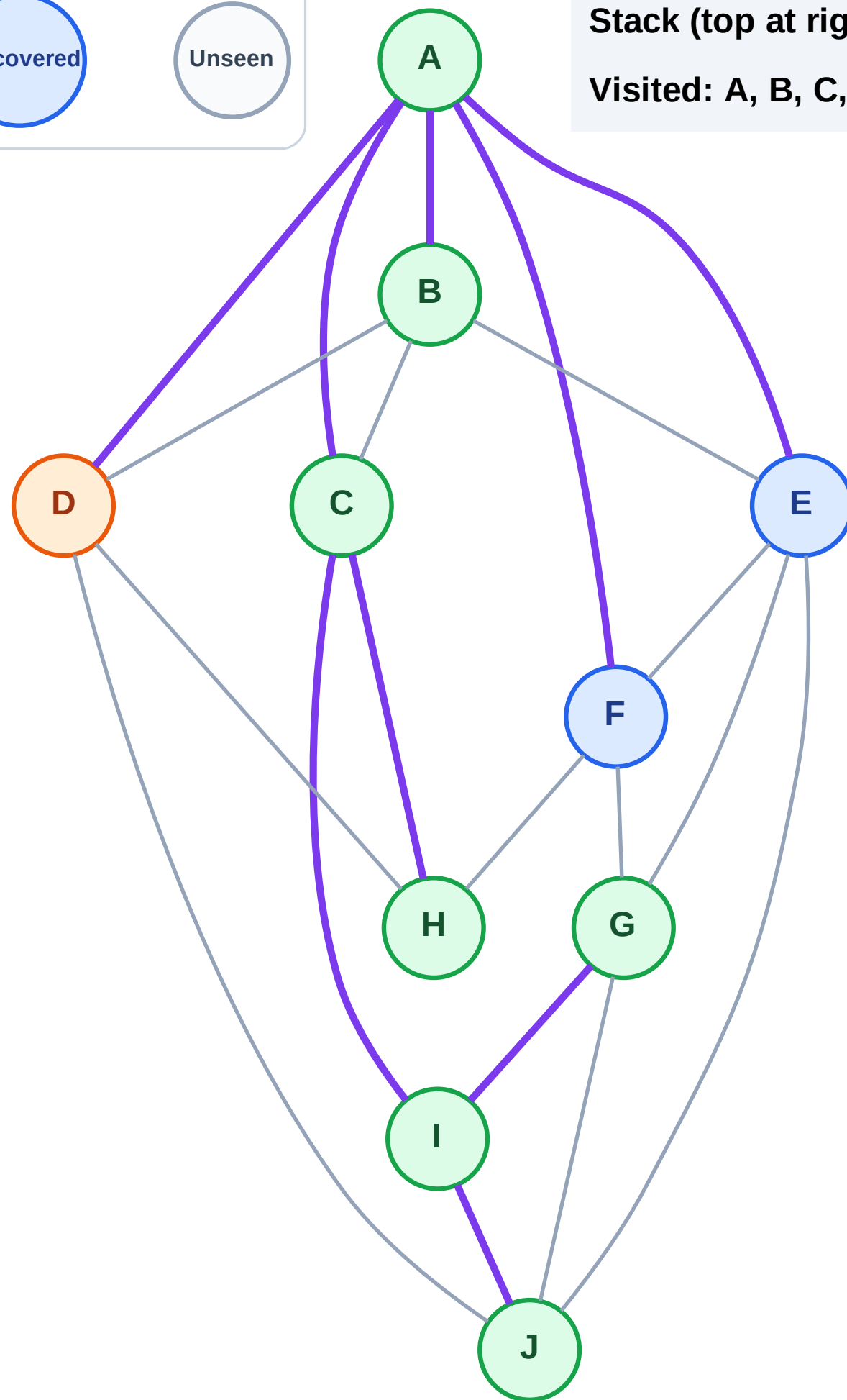
Step 16: Process node D

Legend



Stack (top at right): F | E

Visited: A, B, C, G, H, I, J



DFS Traversal

Step 17: No new neighbors from D

Legend

Complete

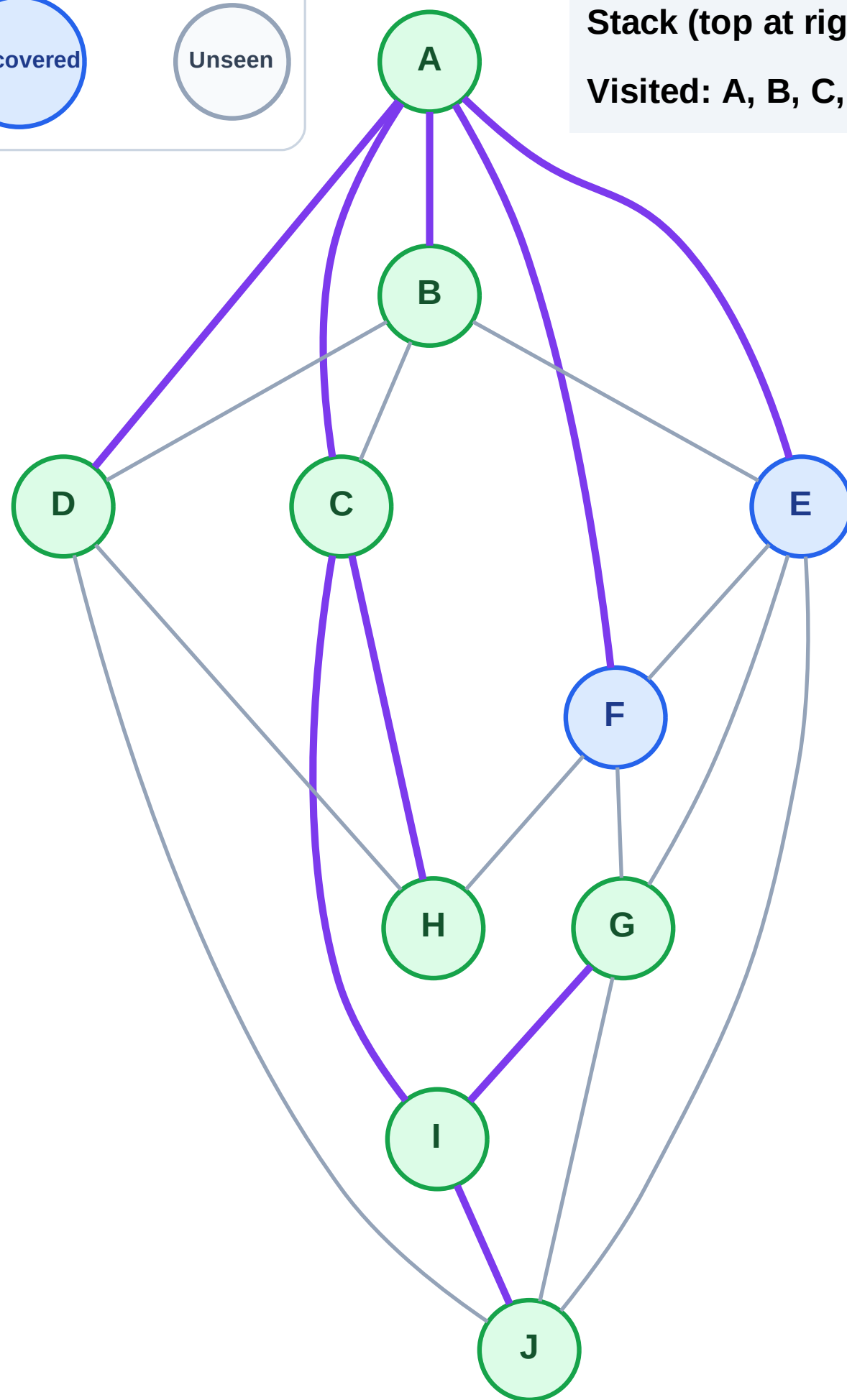
Processing

Discovered

Unseen

Stack (top at right): F | E

Visited: A, B, C, D, G, H, I, J



DFS Traversal

Step 18: Process node E

Legend

Complete

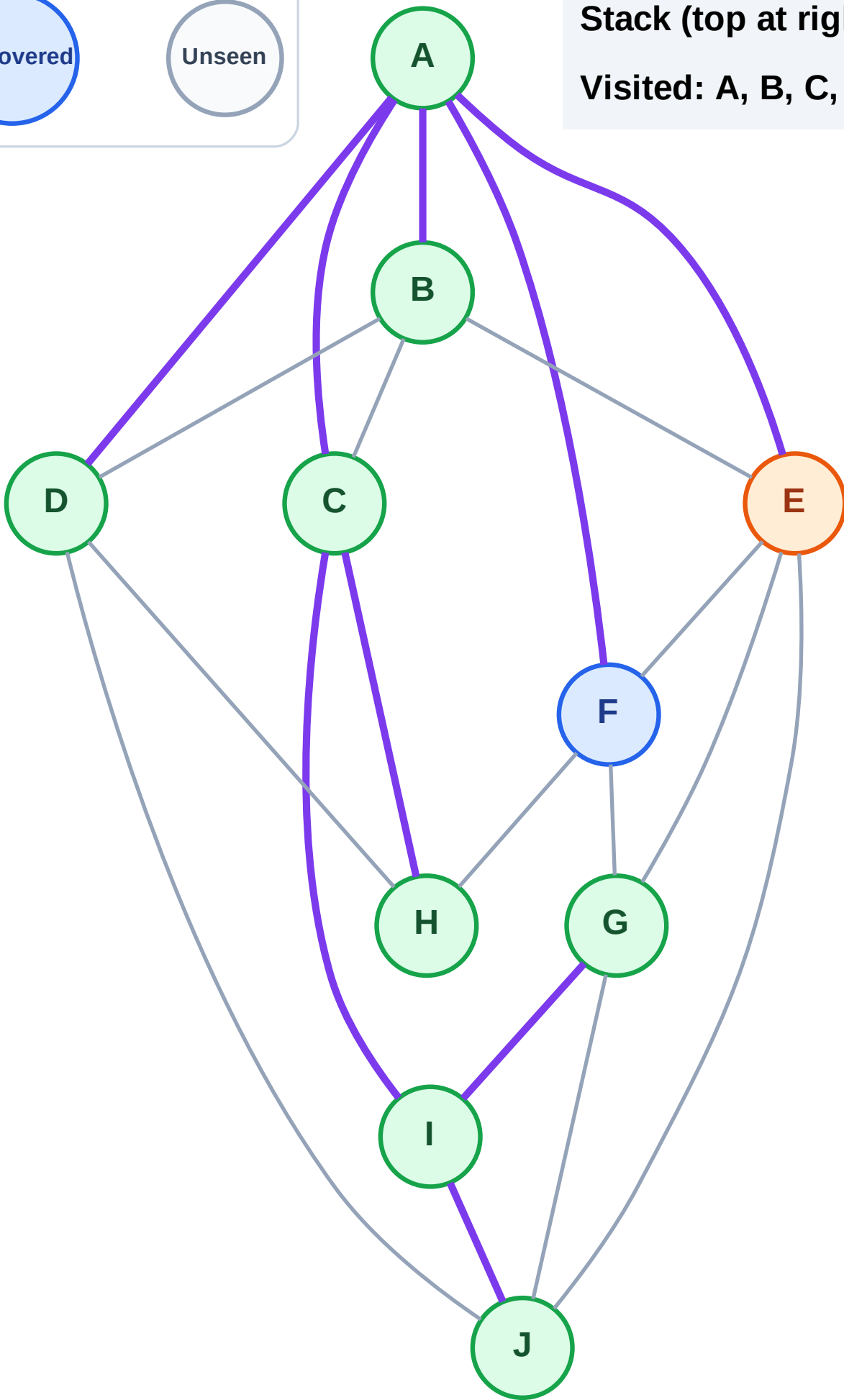
Processing

Discovered

Unseen

Stack (top at right): F

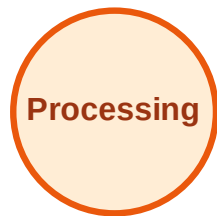
Visited: A, B, C, D, G, H, I, J



DFS Traversal

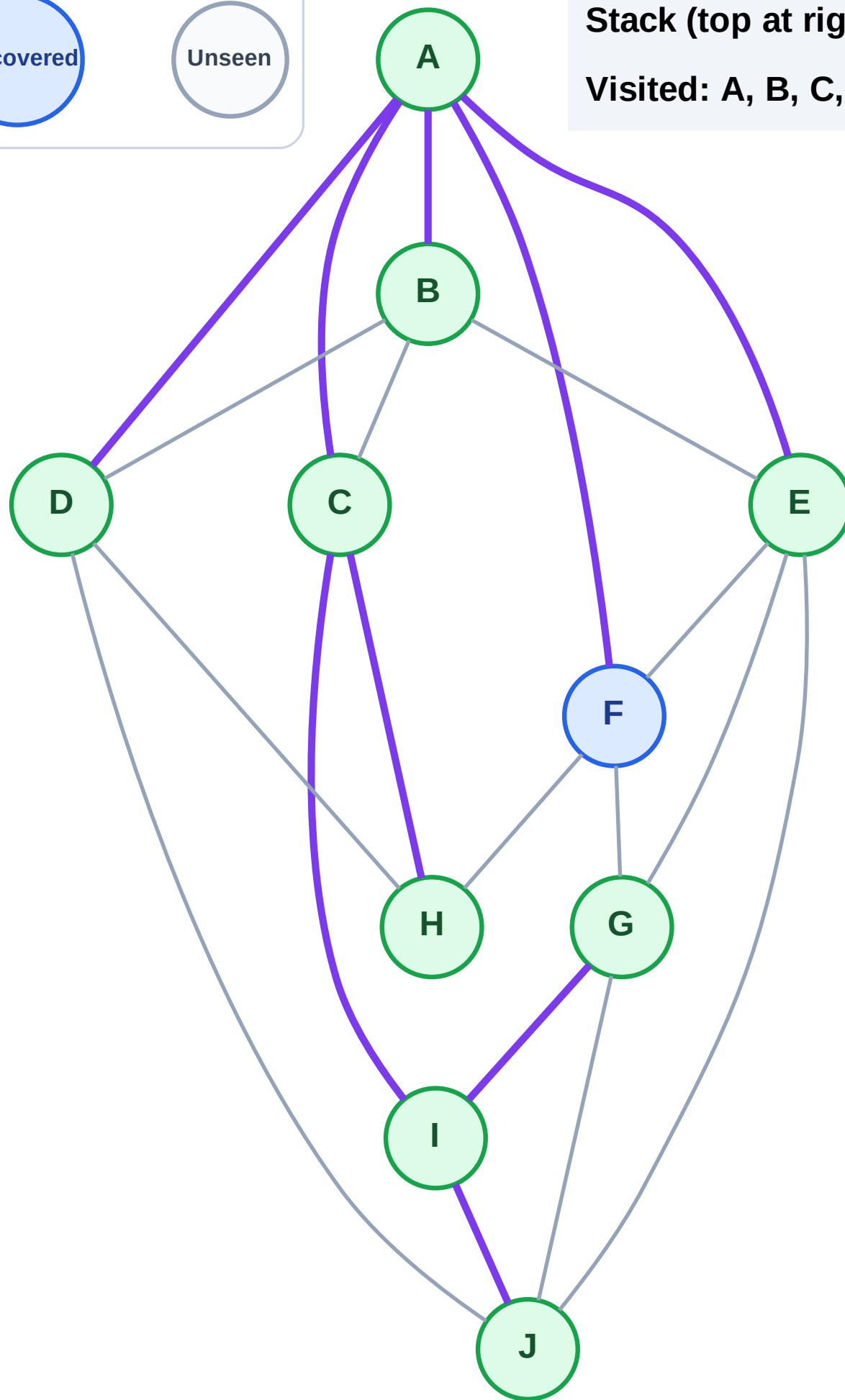
Step 19: No new neighbors from E

Legend



Stack (top at right): F

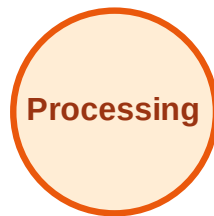
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

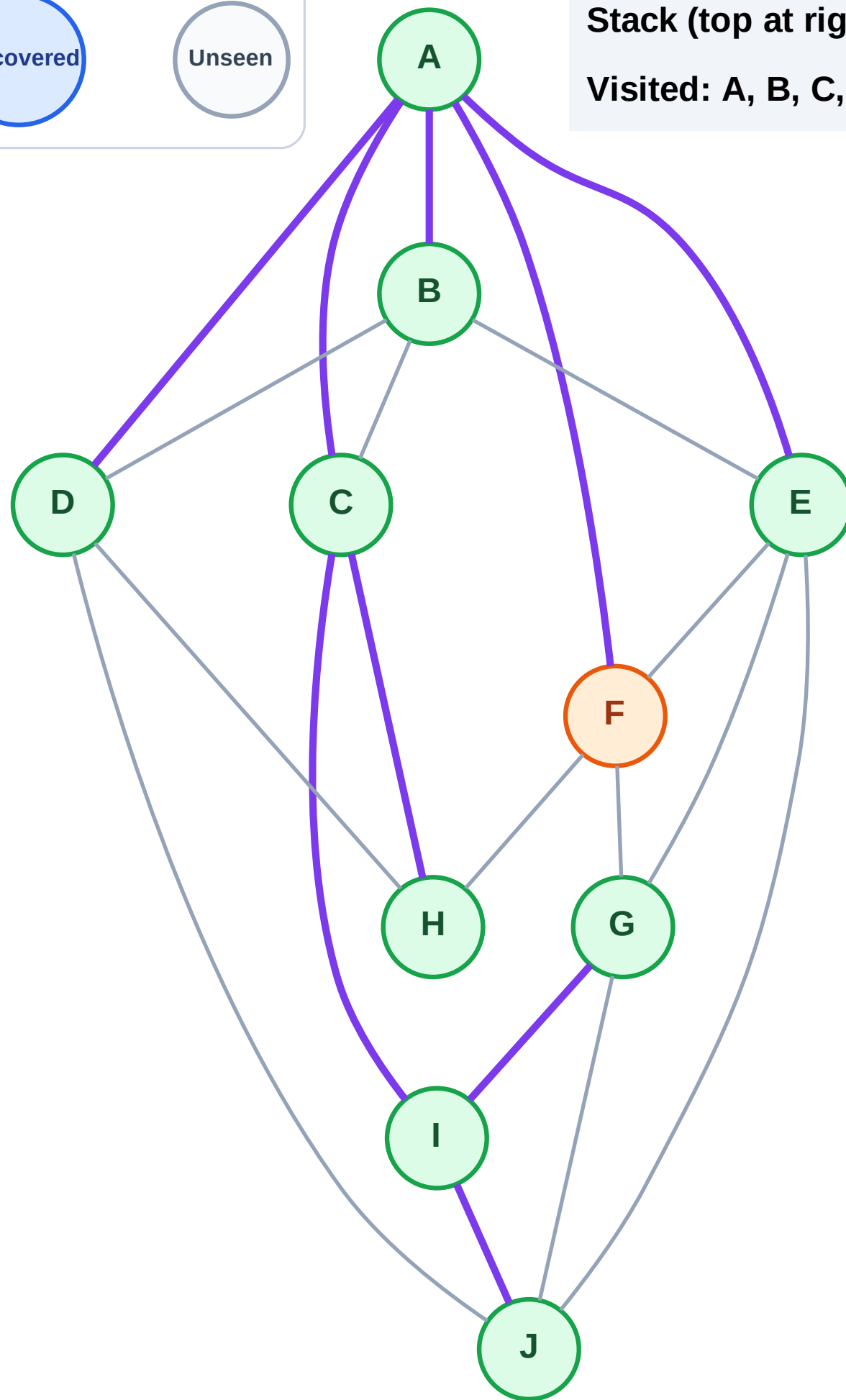
Step 20: Process node F

Legend



Stack (top at right): (empty)

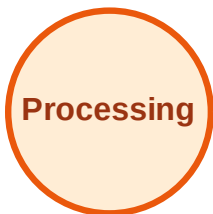
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

